Quantum Advantage

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IBM Quantum

Qiskit Global Summer School 2026

What is Quantum Advantage?

Complexity

Quantum computers solve problems where reliable classical heuristics become computationally unviable. This occurs either due to a fundamental mathematical separation in complexity, or because classical systems simply exhaust practical time and memory resources.

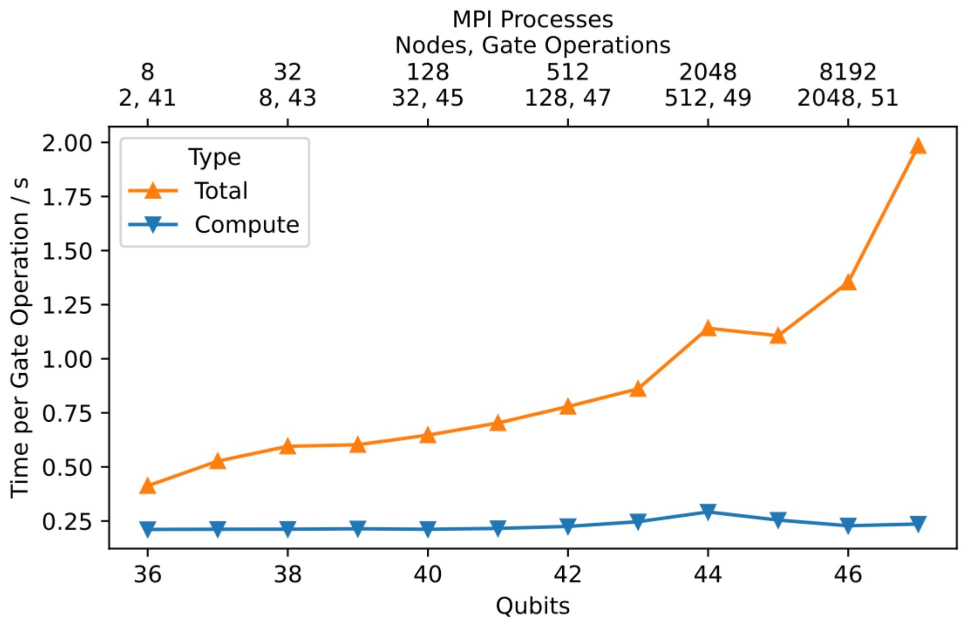
Trust

We must be able to trust the output of quantum machines. When the answers cannot be double-checked classically, we build trust step-by-step through careful calibration and error mitigation, exactly like physicists have done for centuries with telescopes and particle colliders.

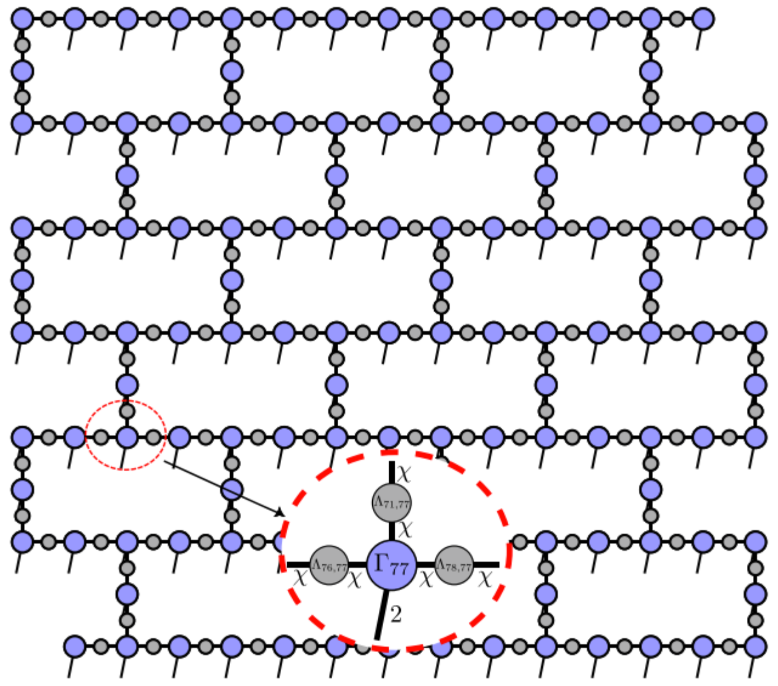
Utility

Moving beyond abstract puzzles designed just to prove hardware speed, the computation must have real-world impact. The results must drive progress in fields like materials science, drug discovery, or fundamental physics.

Classical Heuristics:
Fundamental
Limitations

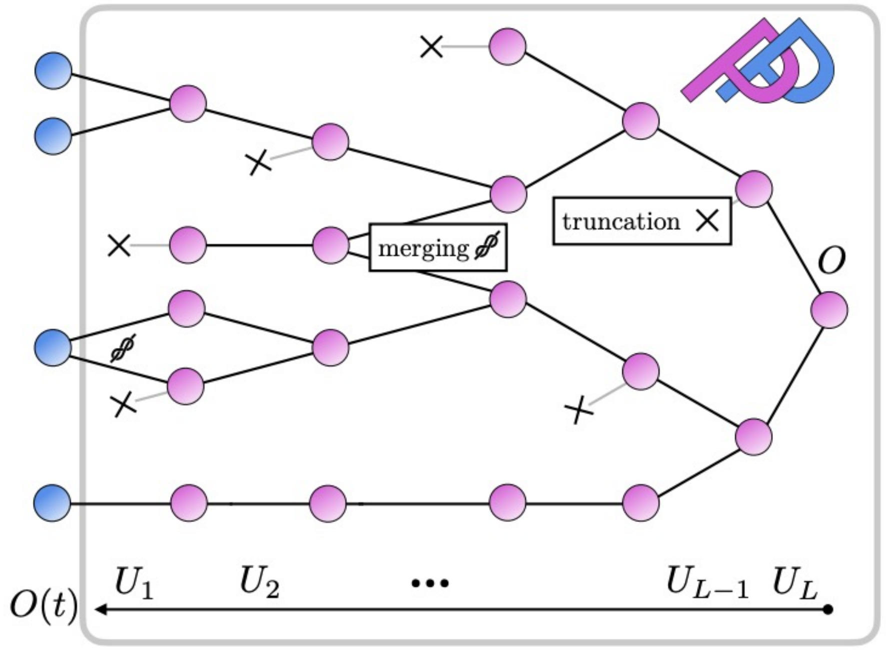


State Vector[1]:
Limited by qubit count



Tensor Networks[2]:
Limited by entanglement

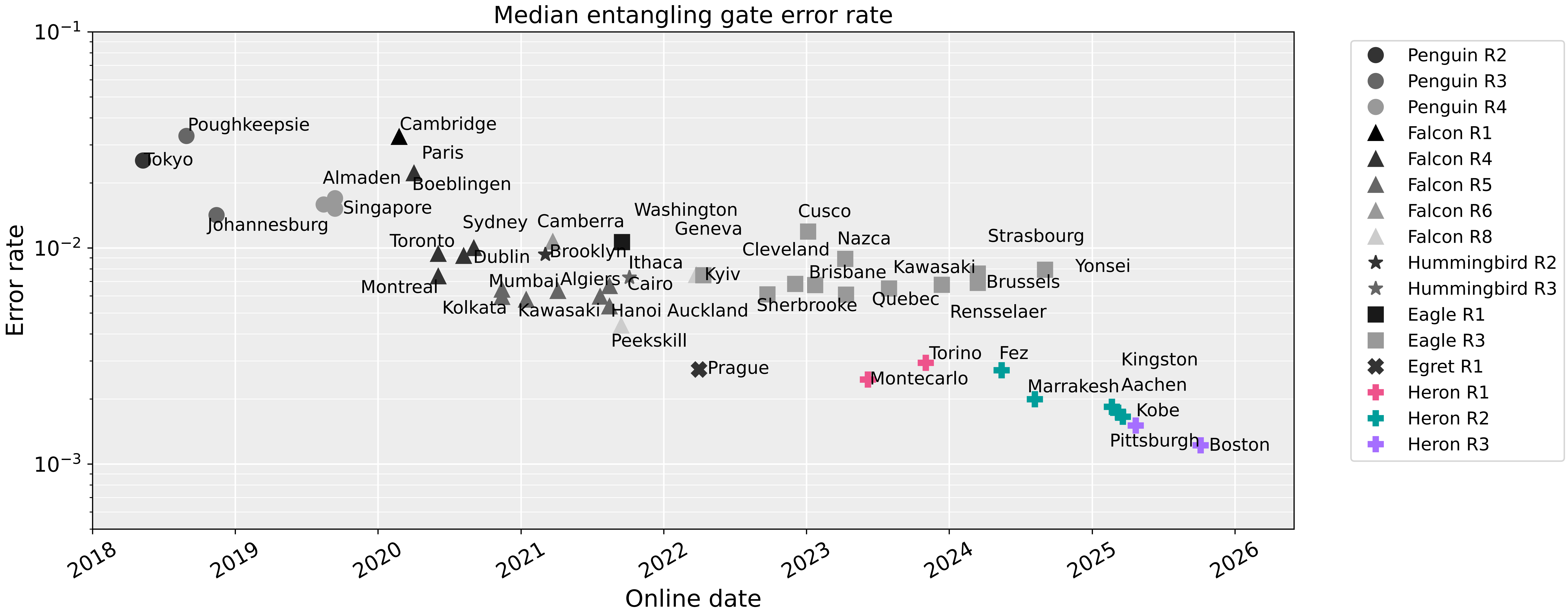
Quantum Devices:
Limited by error rates



Pauli Path[3]:
Limited by operator spreading

Source: 1. arXiv:2511.03359
Source: 2. PRX Quantum 5, 010308 (2024)
Source: 3. Science Advances 10, 3 (2024)

Improving Quantum Hardware

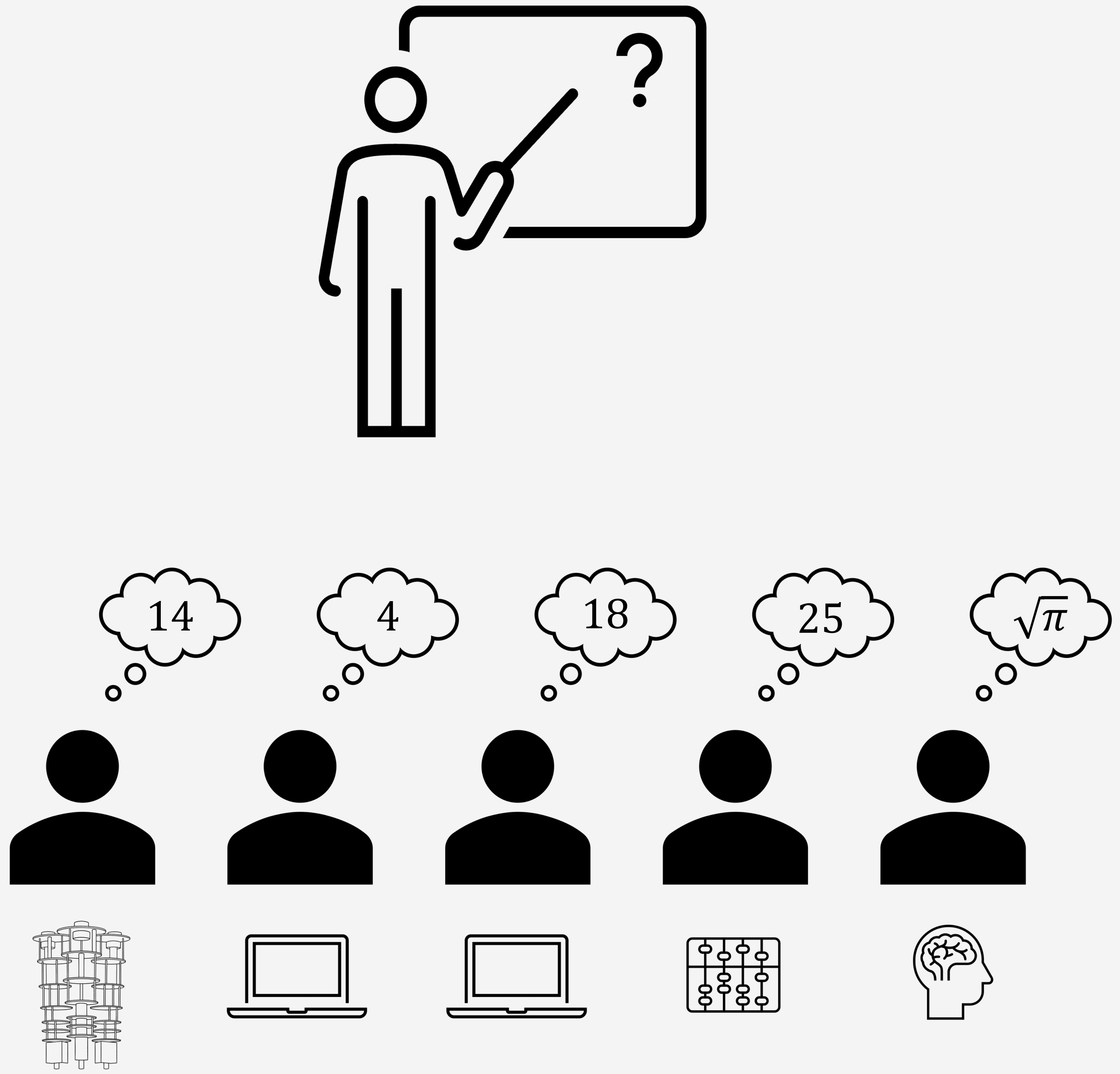


Quantum computers will soon be able to solve problems beyond the capability of classical heuristics

How can we trust any computations?

In this lecture:

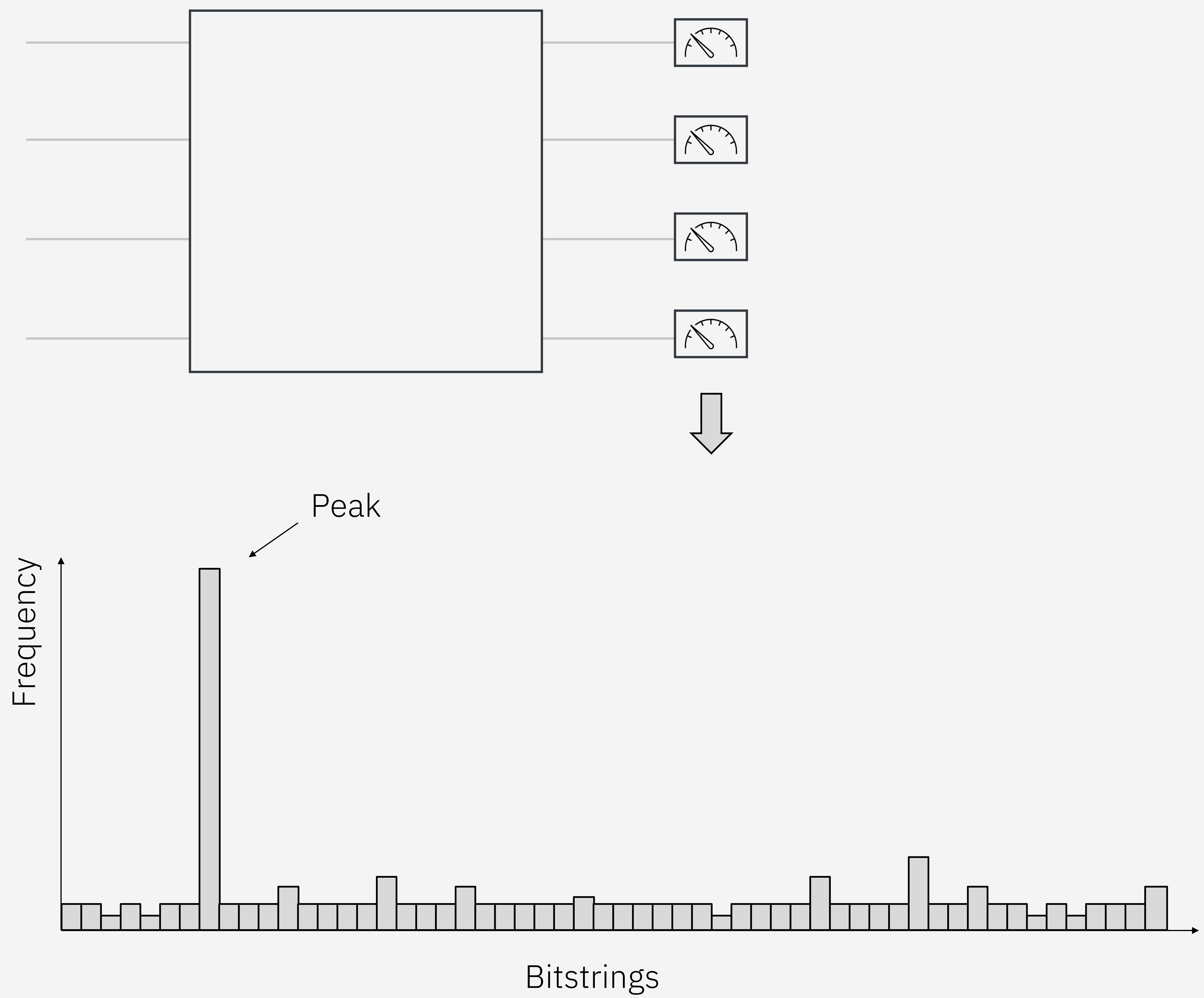
- Example 1: Classically verifiable solutions
- Example 2: Solutions with a built-in quality metrics
- Example 3: Trust-building exercises



Example 1:
Peaked Circuits

Peaked Circuits: Those where the measurement outcomes are peaked around one bitstring.

The peak bitstring can be efficiently verified classically (e.g. Shor algorithm) or known by design[1,2].



Source: 1. arXiv:2404.14493
Source: 2. arXiv:2510.25838

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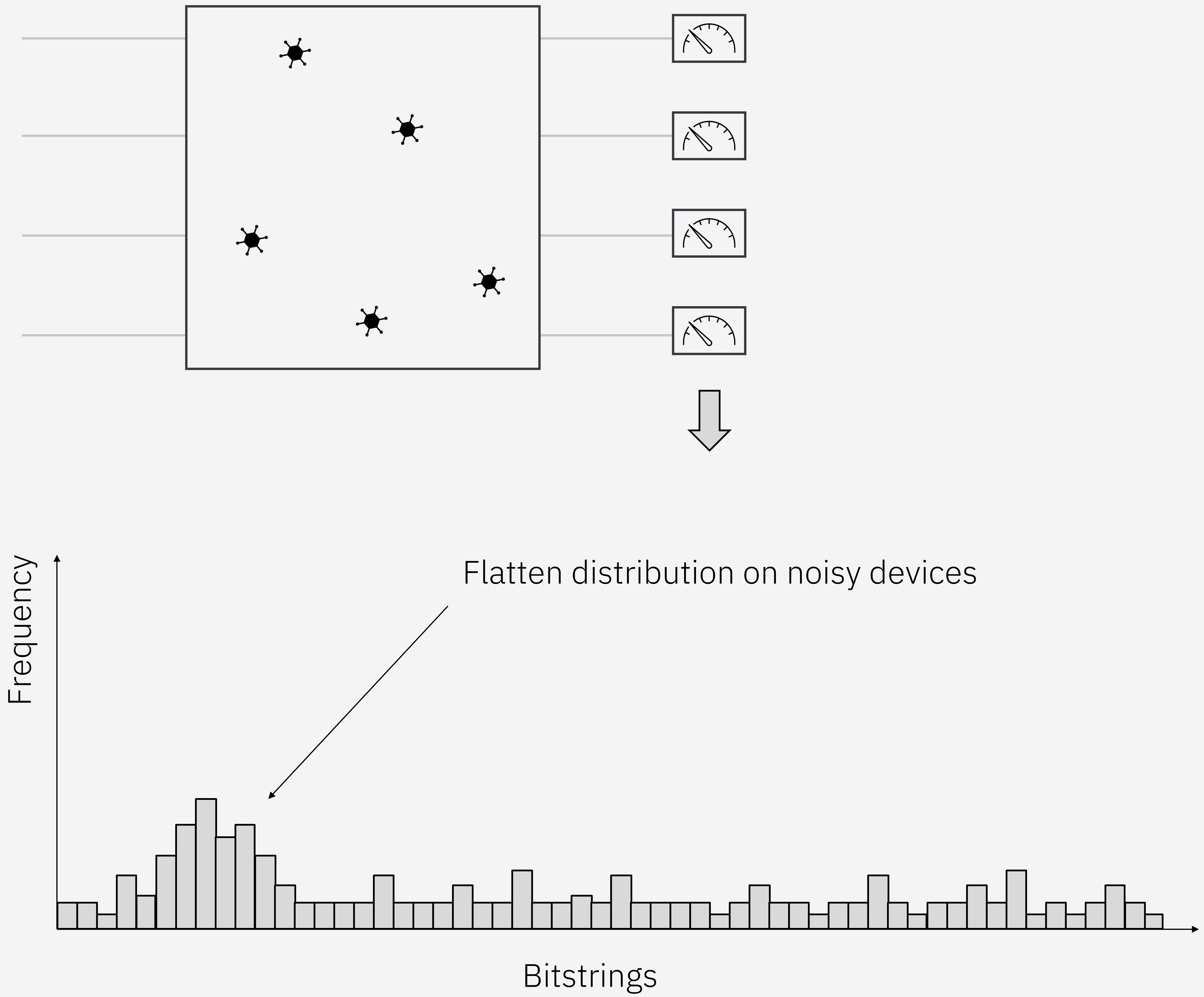
Challenge: Distill the peak from a noisy, flatten distribution



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Example: Majority voting

Samples from a noisy device

'111000011110100110 10...1000'
'111101011110100110 10...1000'
'111111111110100110 10...1000'
'100000010011011101 01...0010'
'000111011101000100 01...1011'
'111000010000011100 00...1100'
'110011101111001011 00...0100'
⋮
'011000000010101010 00...0011'

Distilled bitstrings

'111000011.....................................'

Source: 1. arXiv:2404.14493
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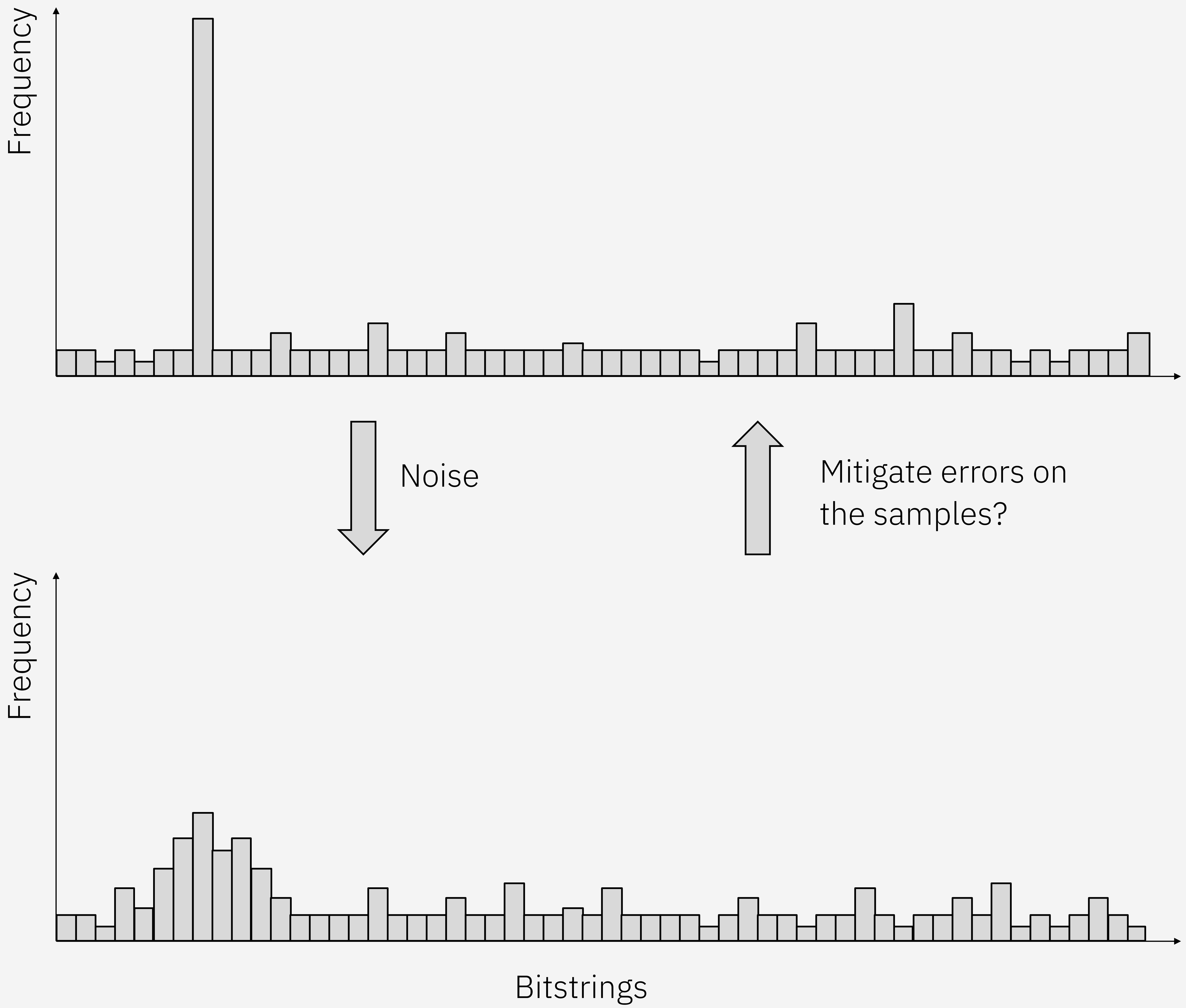
Error mitigation on quantum samples is an active area of research



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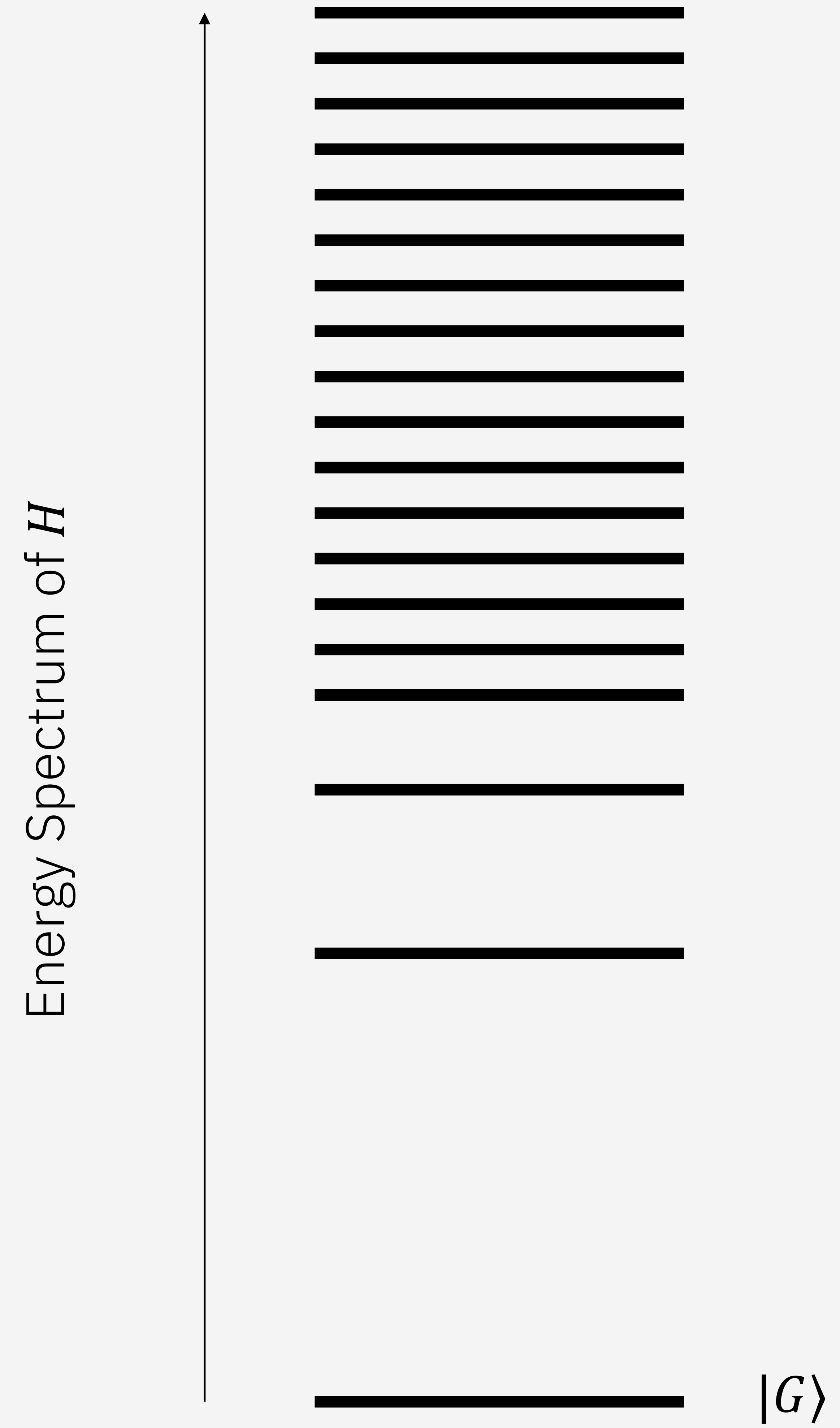
Example 2:
Ground State Approximation

Problem: Approximate the ground state $|G\rangle$ of a given Hamiltonian H.

An important task in both physics, chemistry, and material science.



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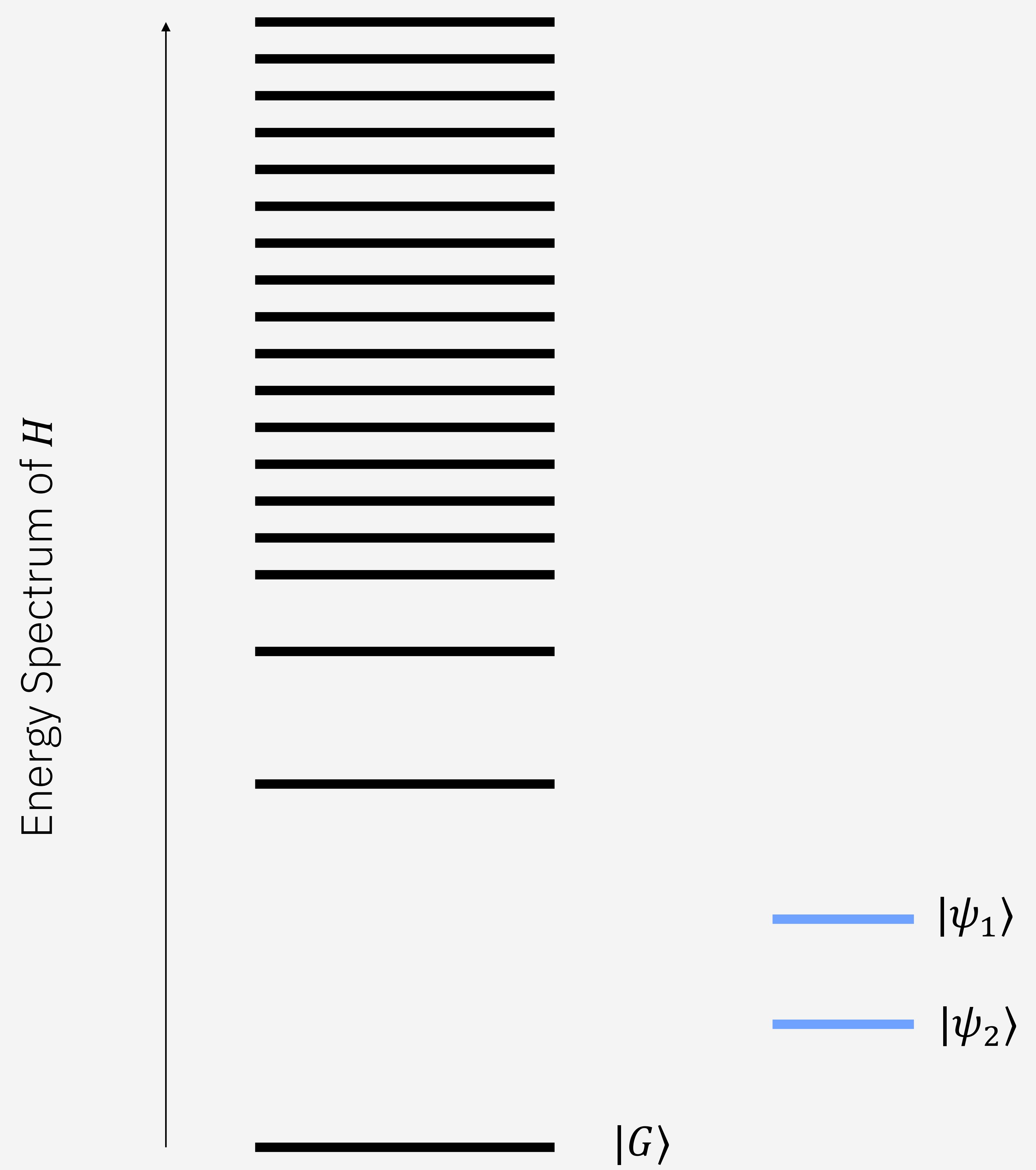
Built-in quality metric: The lower the energy $E = \langle\psi|H|\psi\rangle$ is, the closer $|\psi\rangle$ is to $|G\rangle$.



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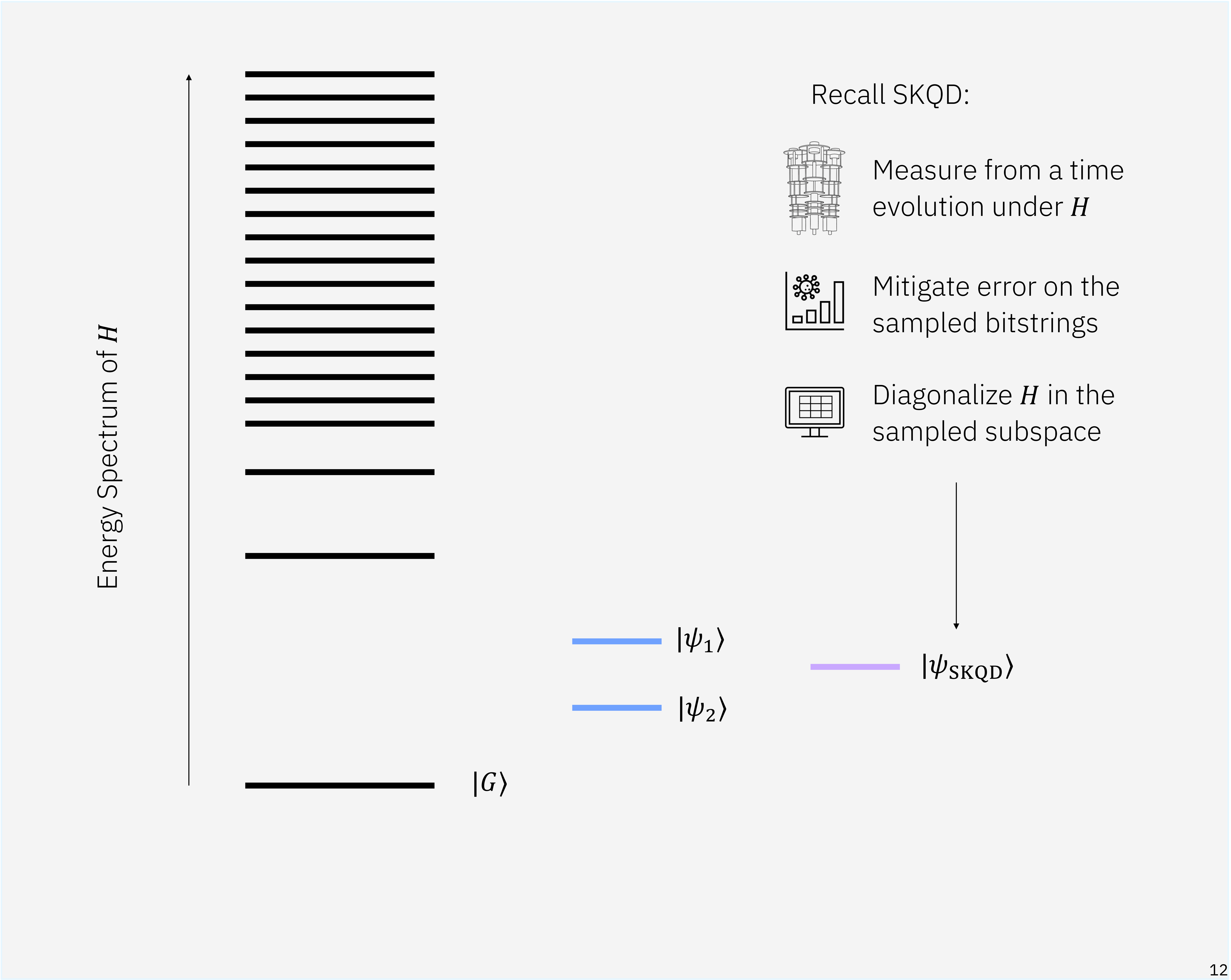
Quantum heuristics, e.g. SKQD, are becoming more competitive.



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Observable Estimation

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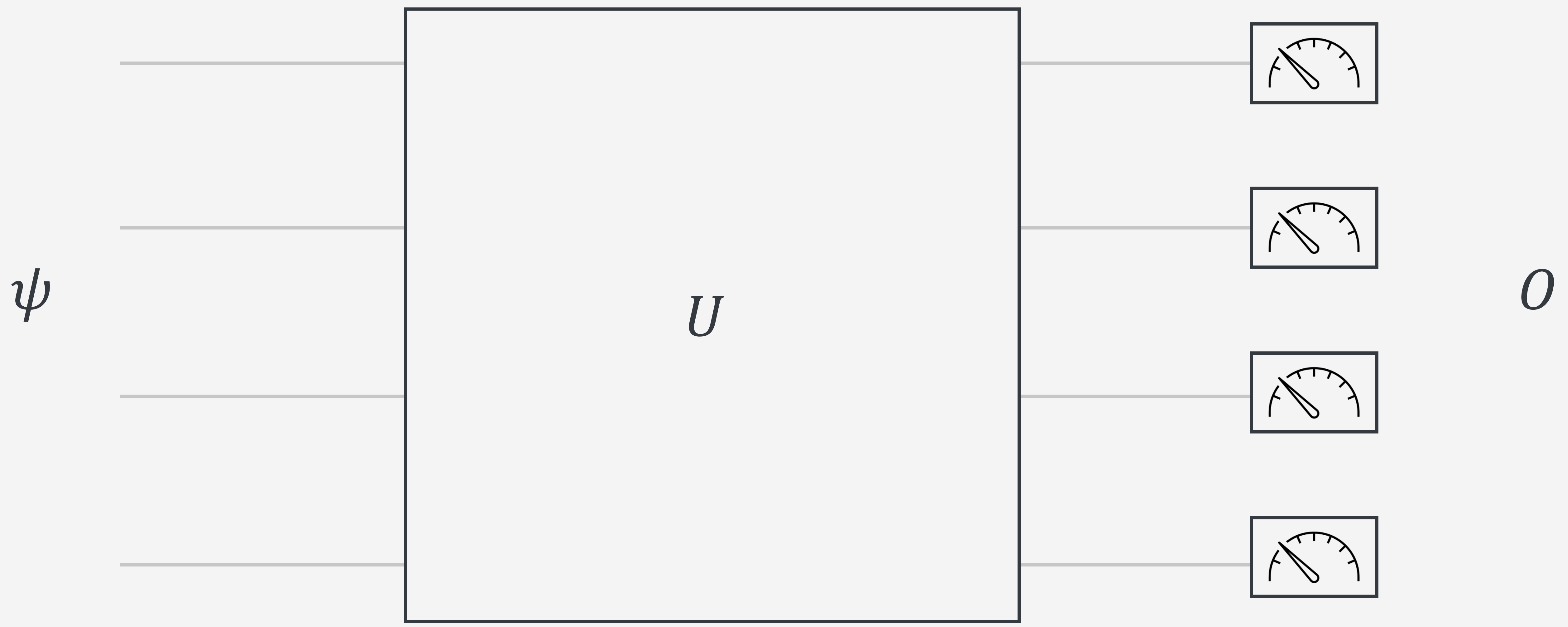

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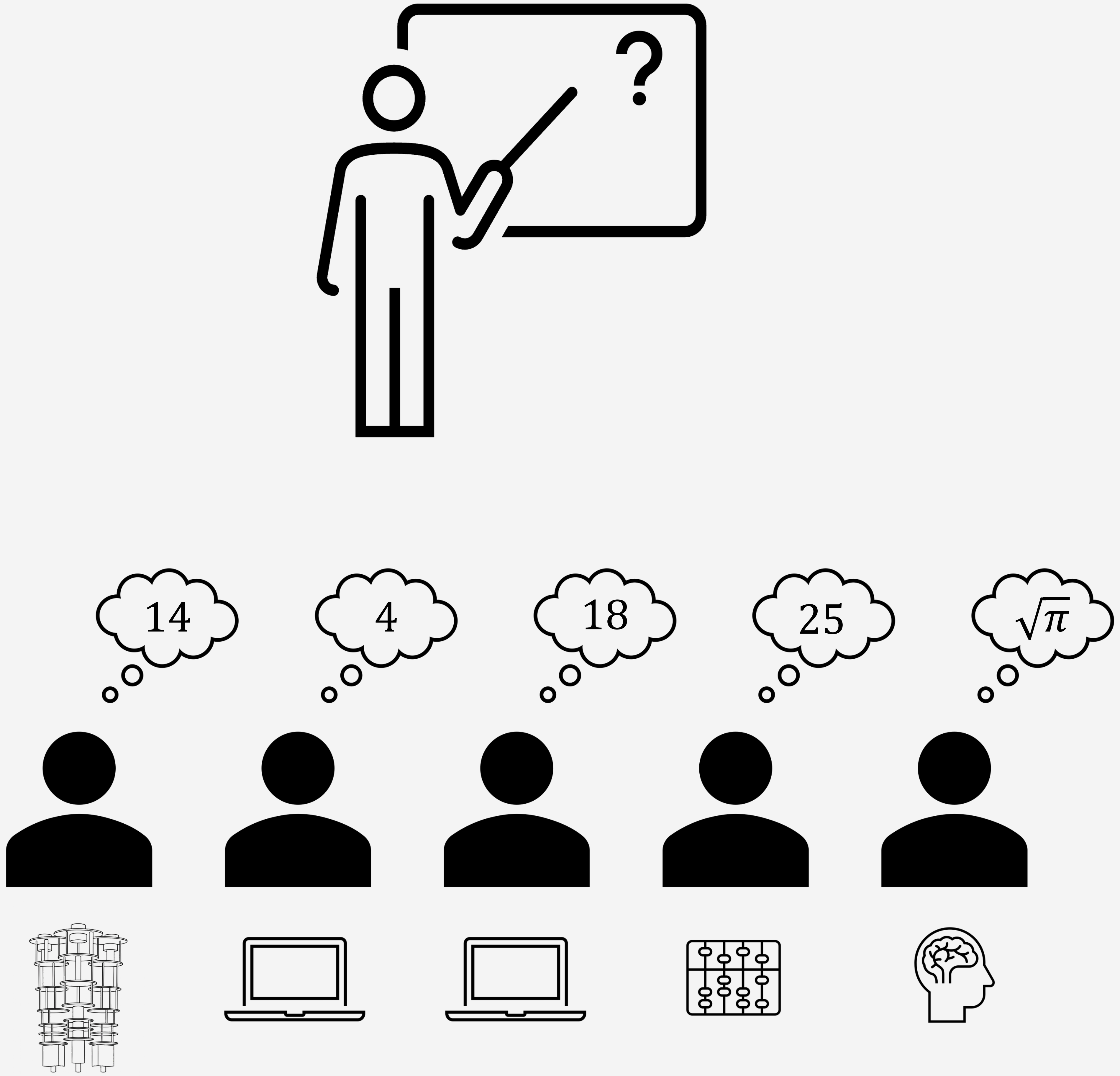

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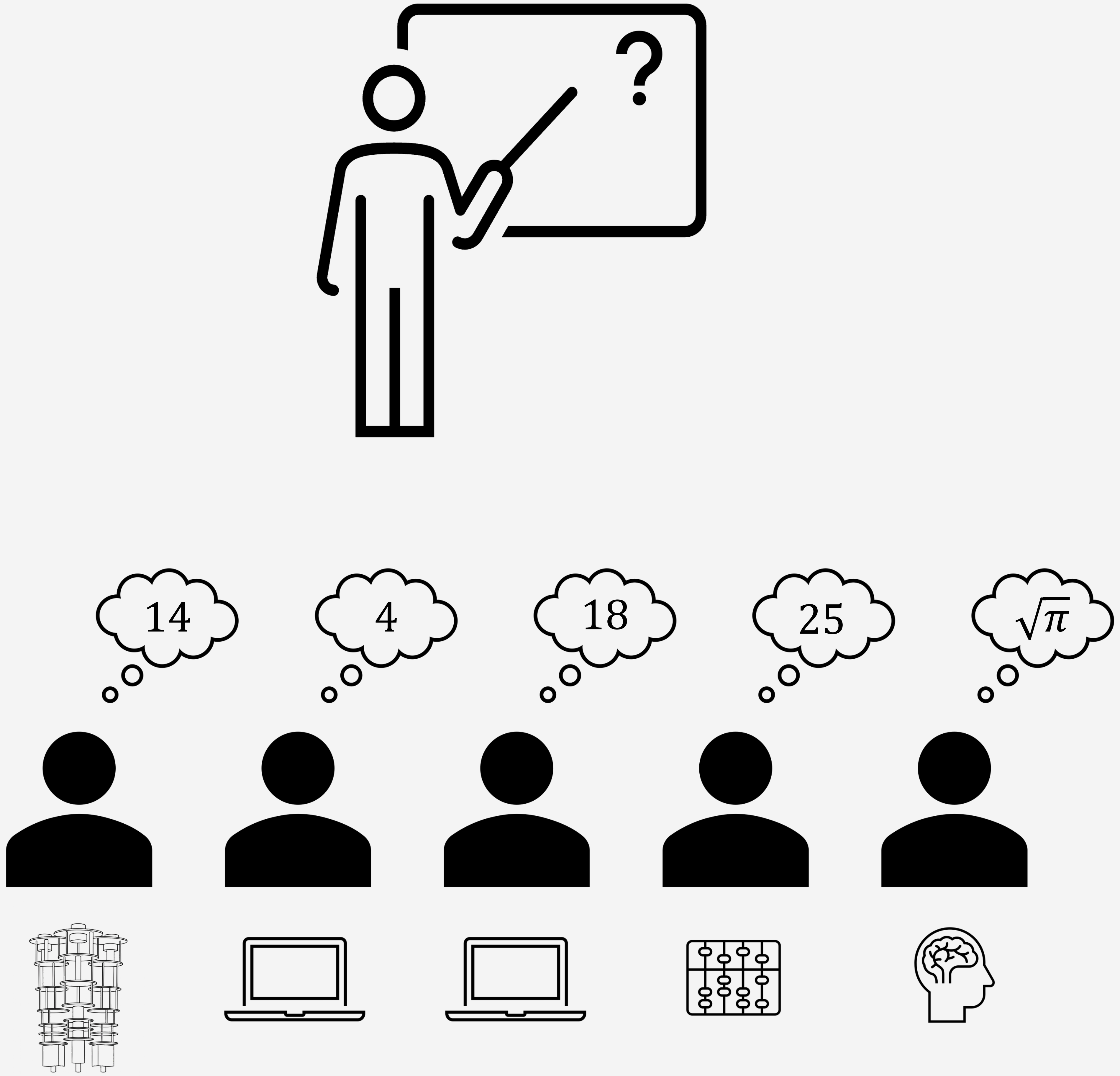
Not a new challenge!

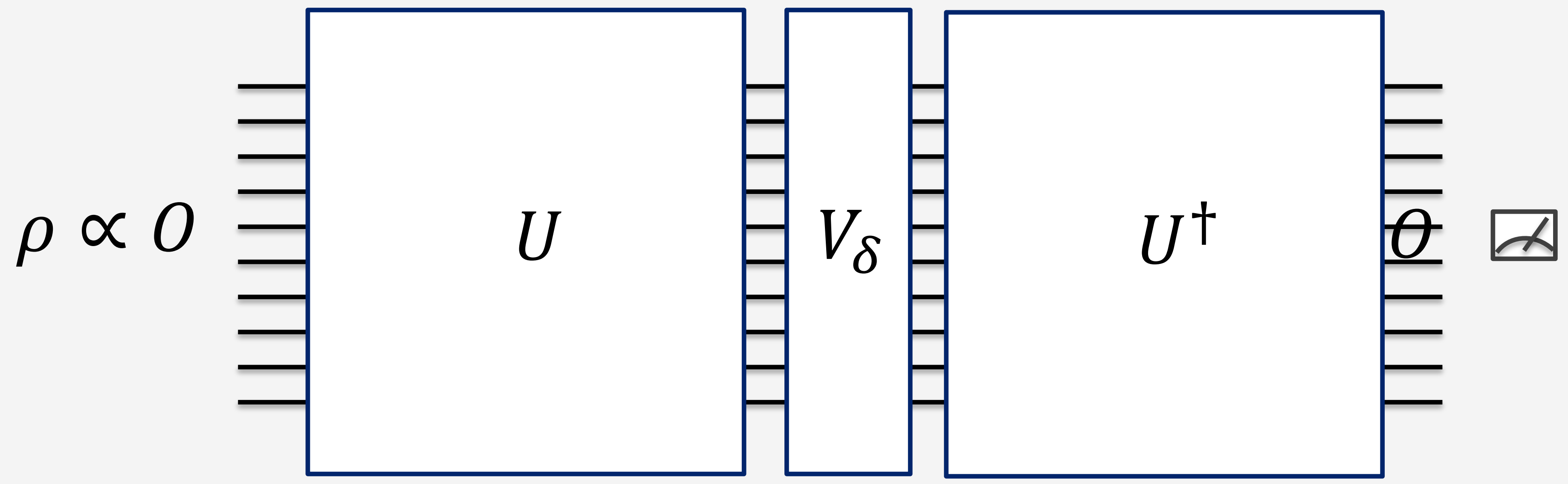
Conventional approach: collect a lot of evidences:

- Benchmark on small problems where exact solutions are known
- Reproducibility under different conditions
- Theoretical predictions
- ...



Example 3:
Observable Estimation

Example: Operator Loschmidt Echo



Signal $\approx 1 - \frac{1}{2}\delta^2 \times \mathrm{OTOC}$

$\Rightarrow$ Probe operator spreading

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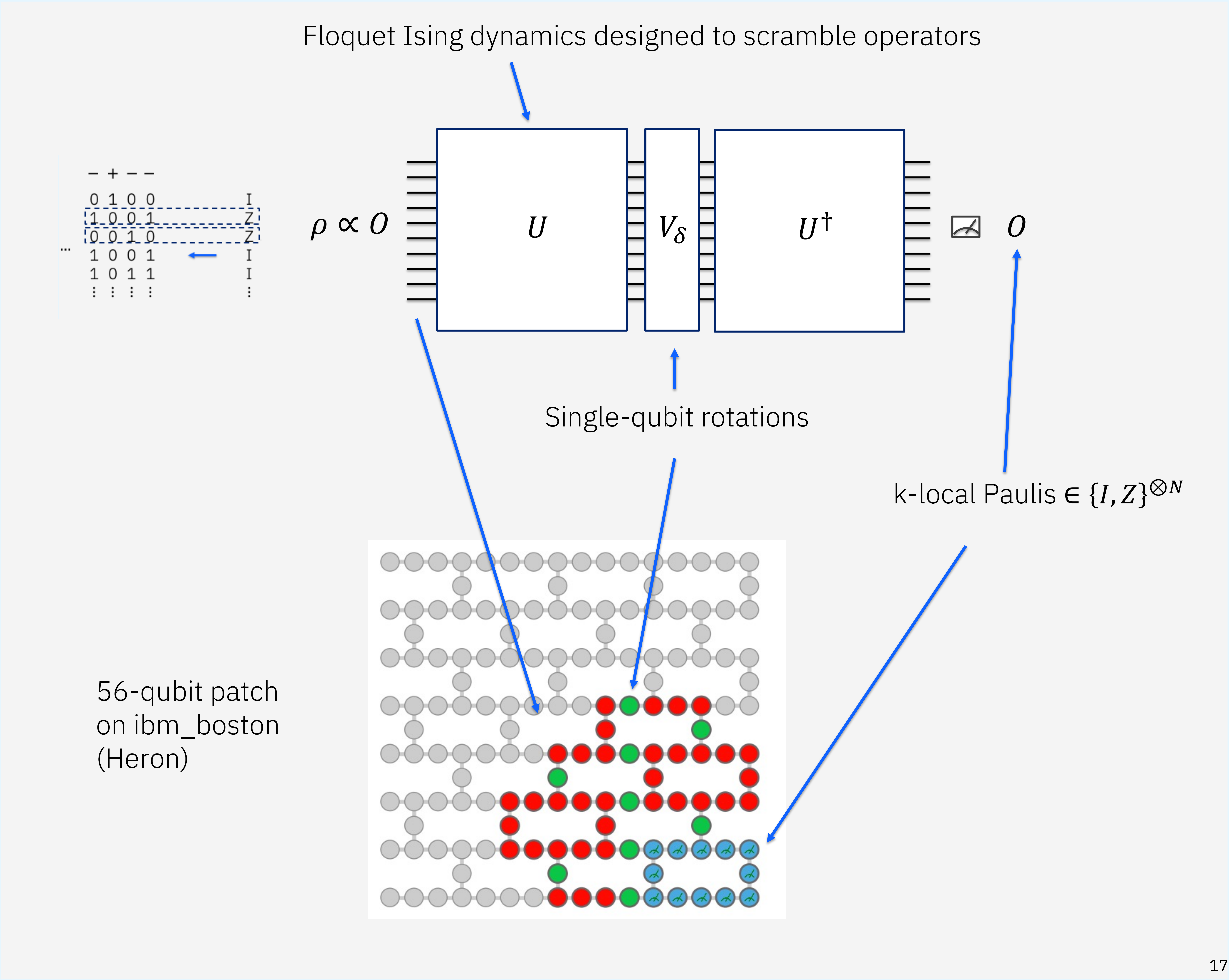

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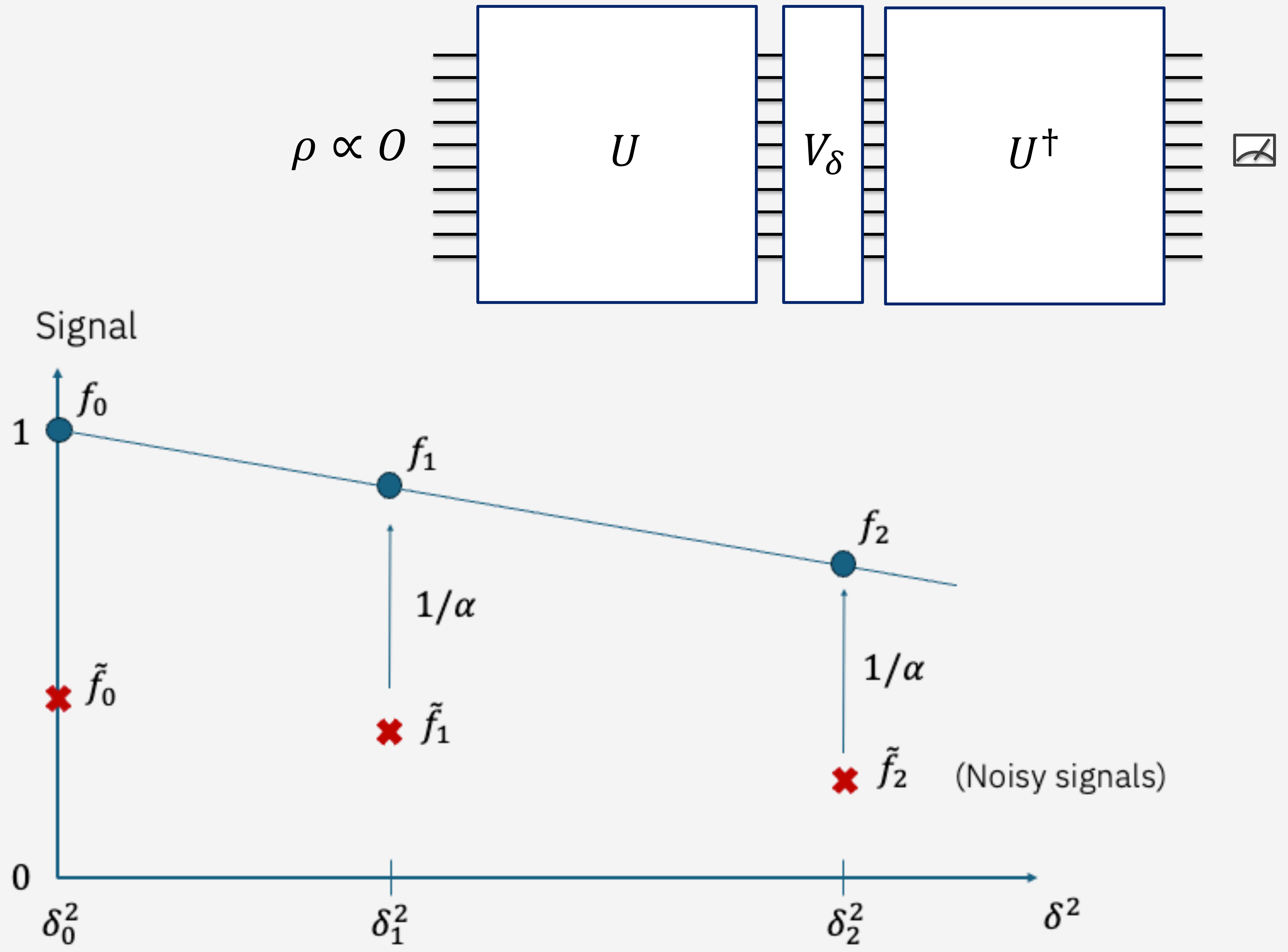
Heuristic error mitigation: Global Rescaling



Assumption: decay factor $\alpha = \dfrac{\tilde{f}_0}{f_0} = \dfrac{\tilde{f}_1}{f_1} = \dfrac{\tilde{f}_2}{f_2}$ independent of δ

$\Rightarrow$ Rescale $\tilde{f}_1$ by $1/\alpha$ to get f_1

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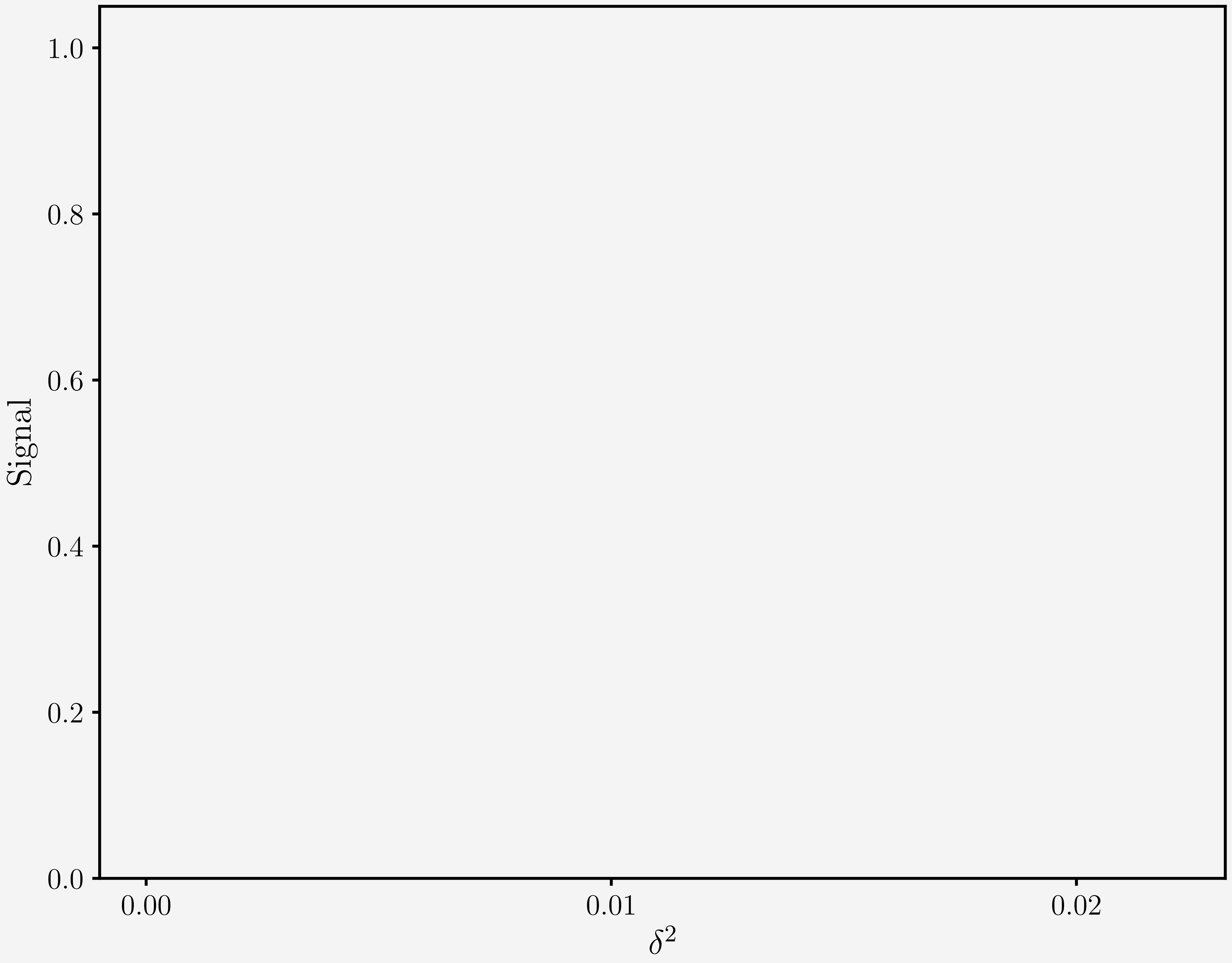
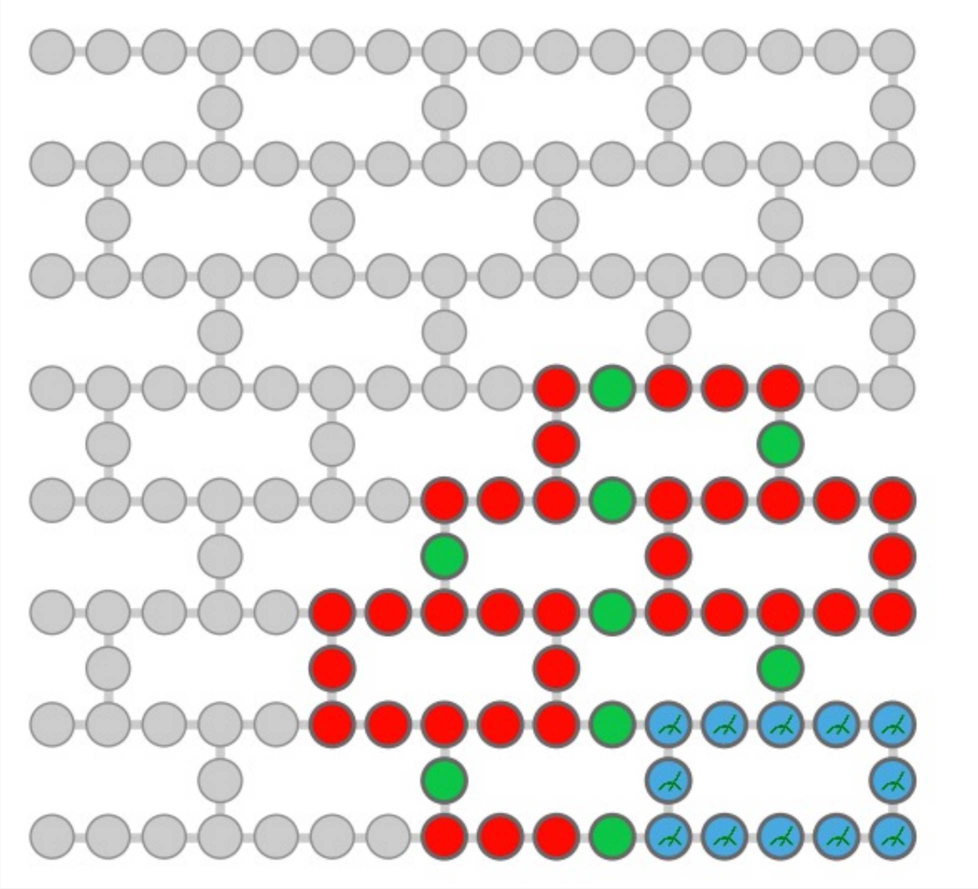
OLE at Depth 72 (1488 CZ gates)



ibm_boston

56 qubits,
depth 72 (1488 CZ gates)



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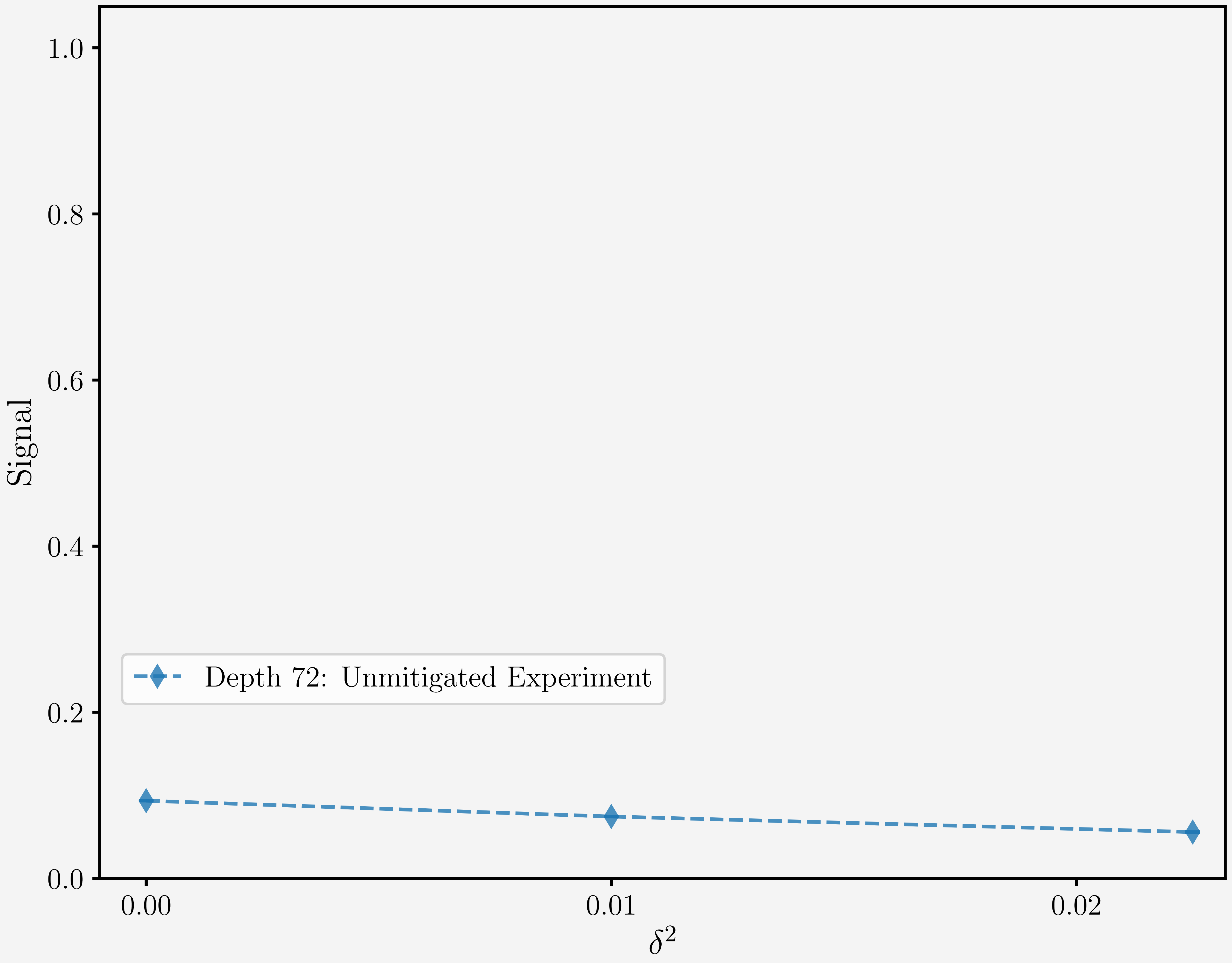

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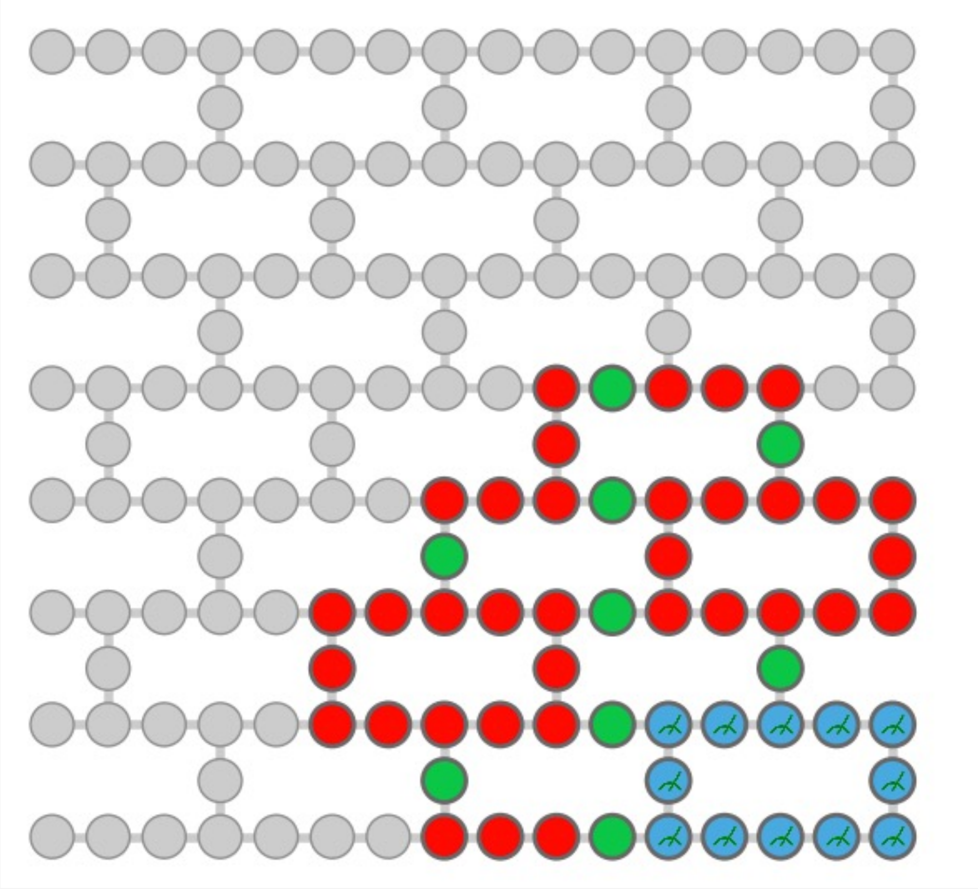

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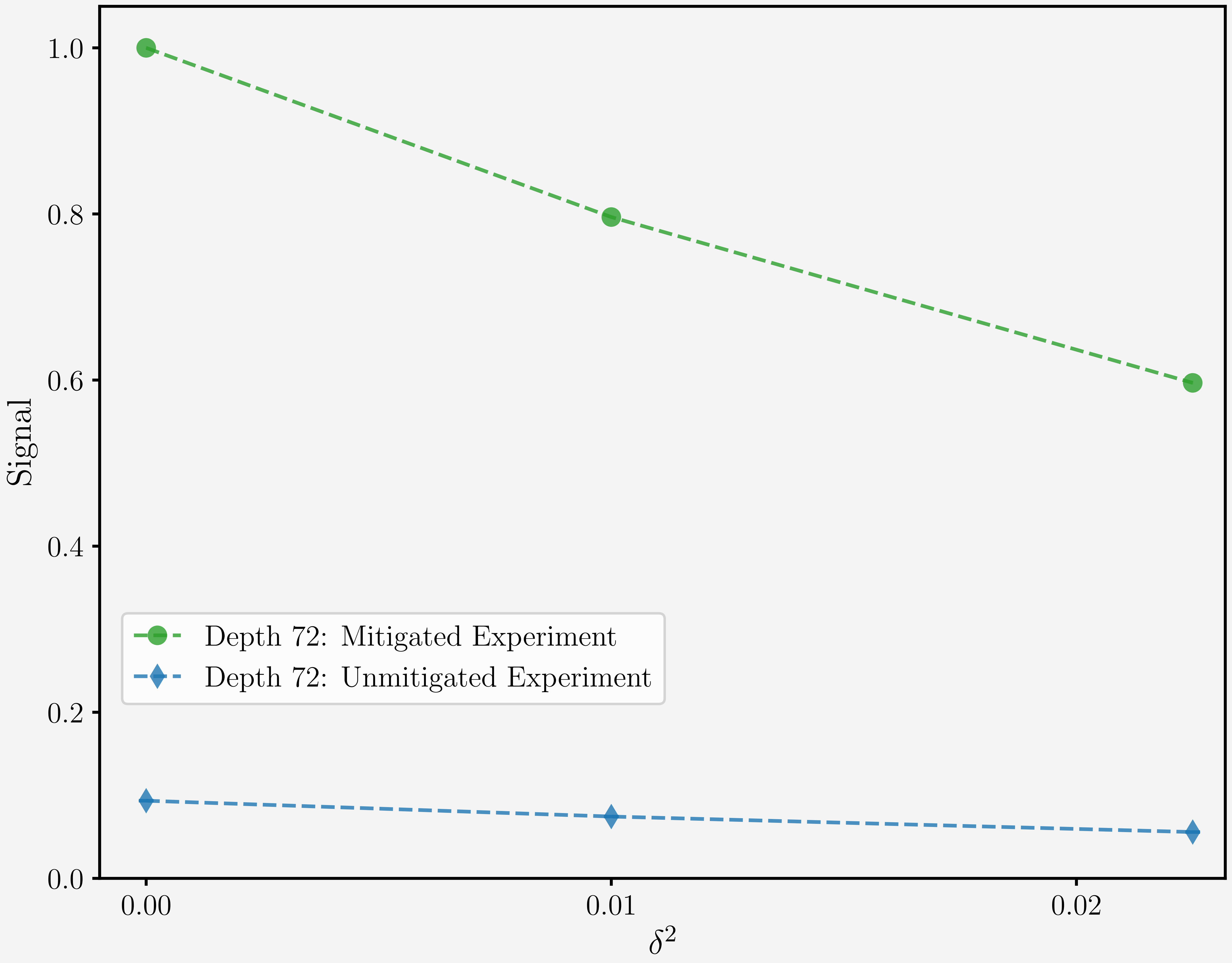

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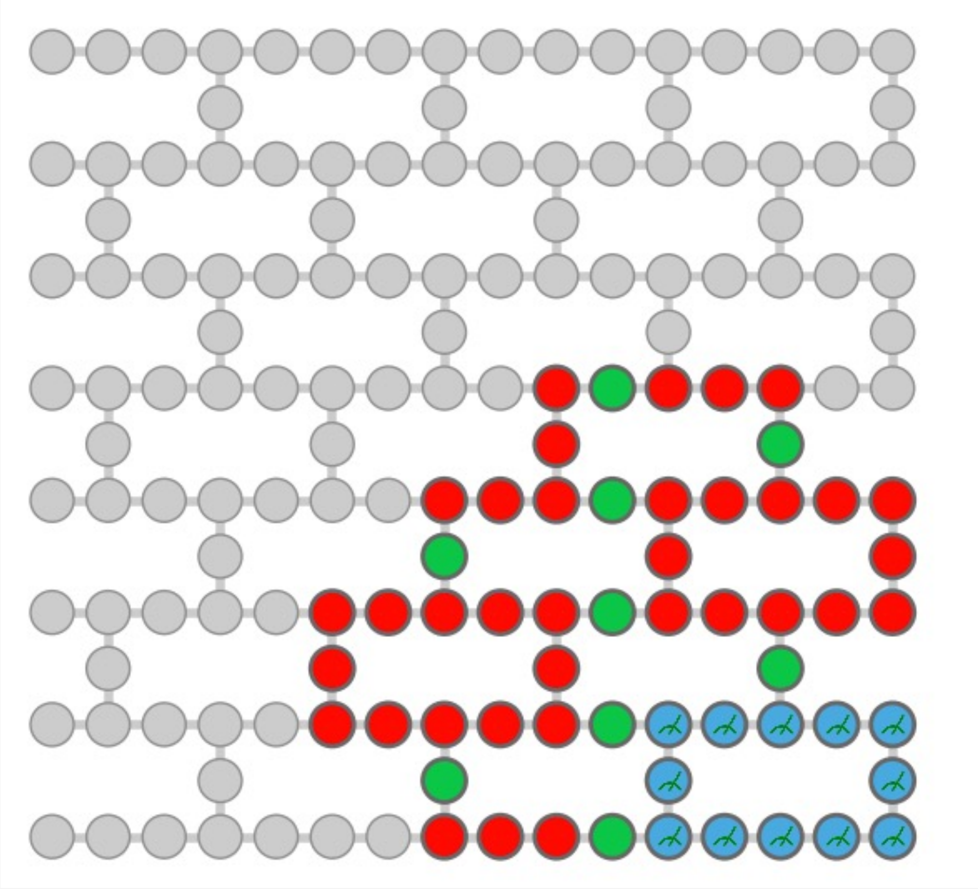

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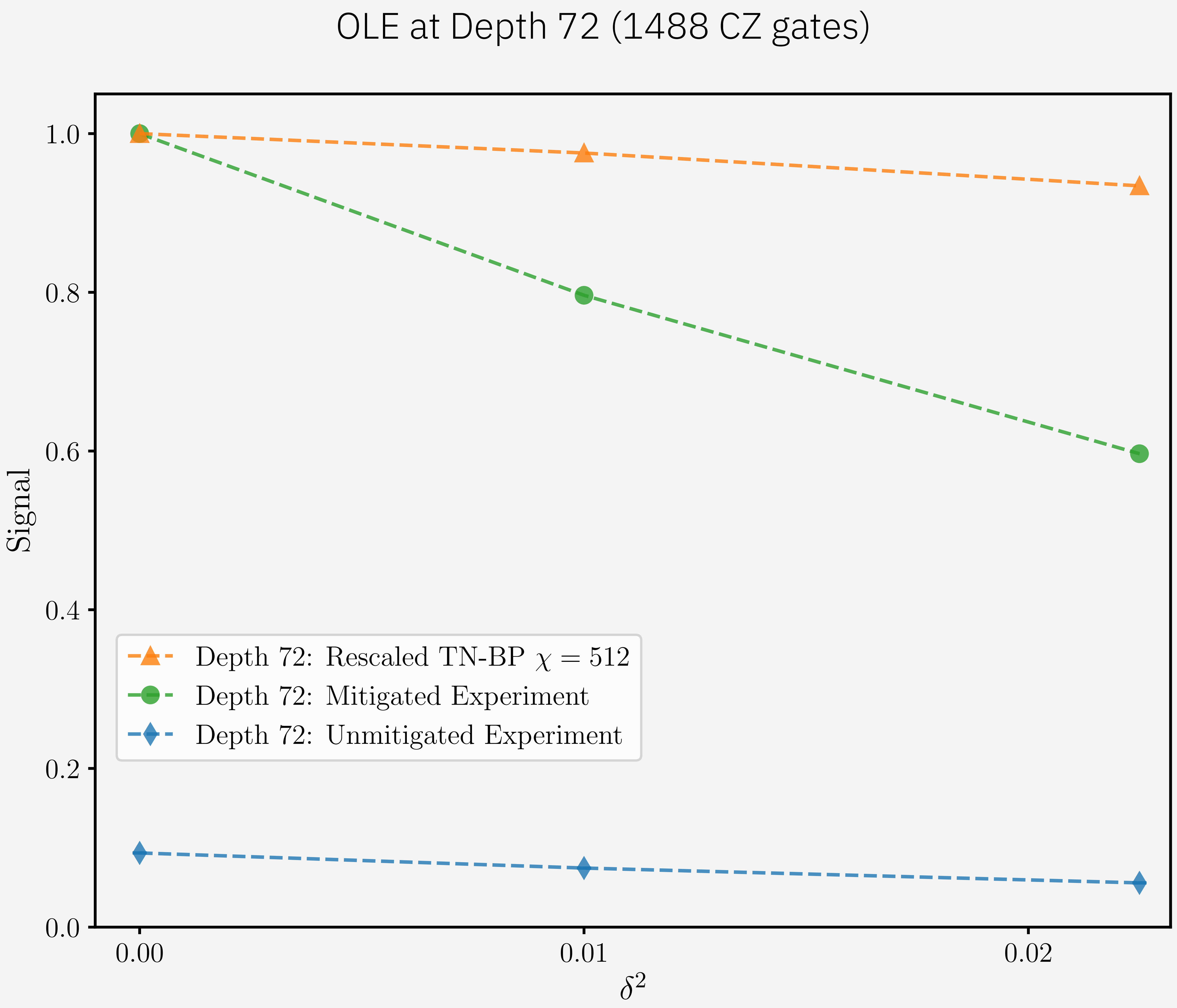

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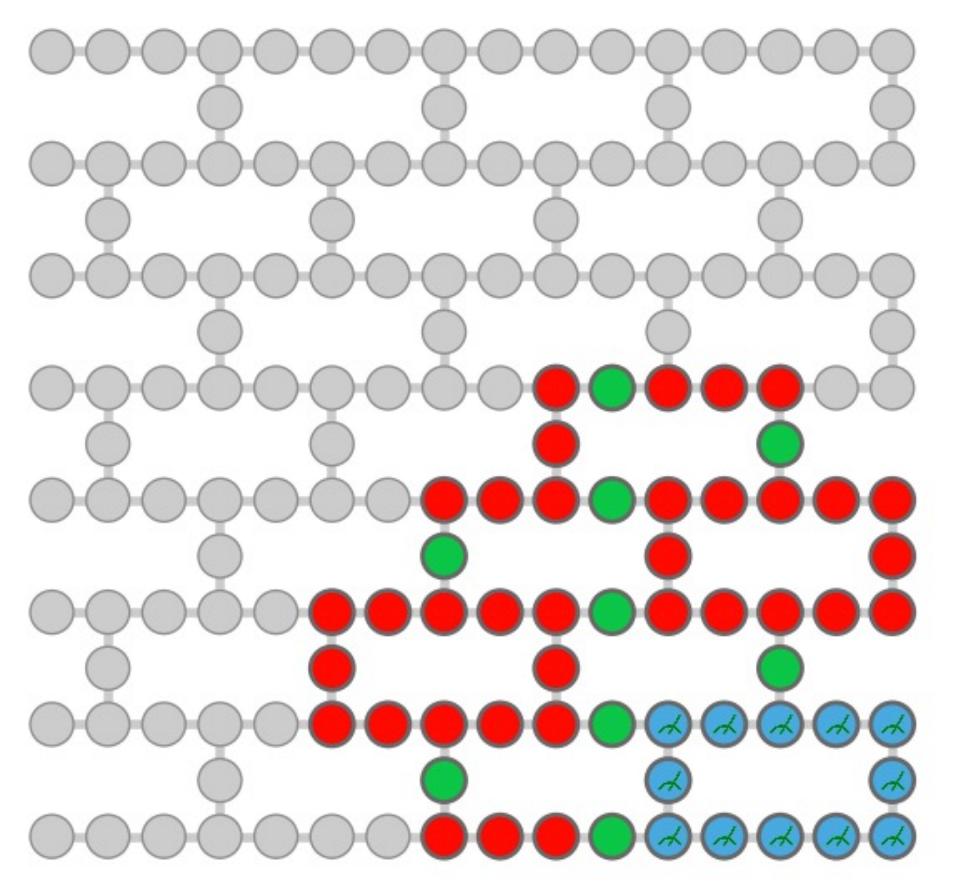
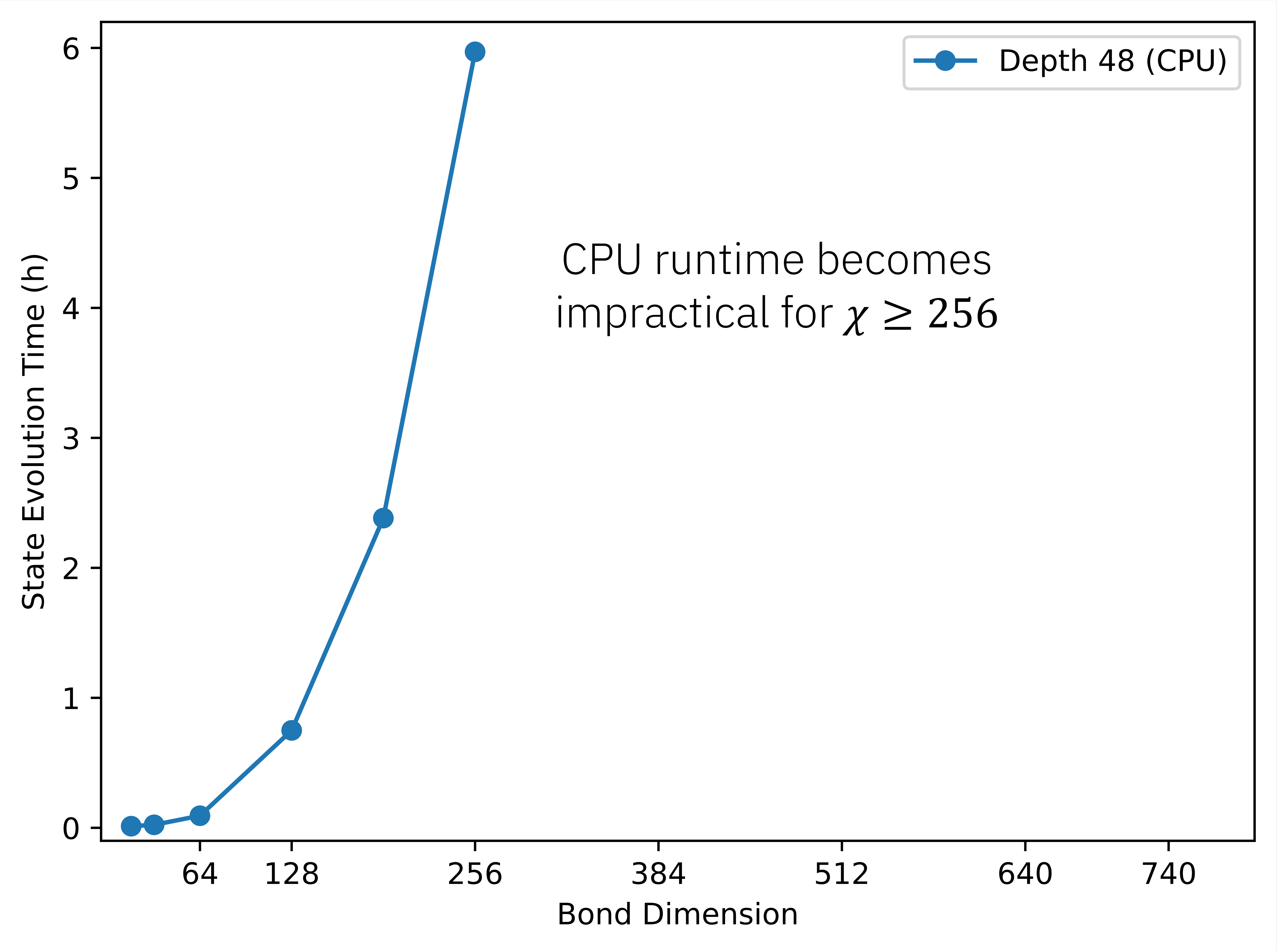


Example 3:
Observable Estimation

Increase resources for TN-BP and hope for convergence?

TN-BP runtime for OLE at Depth 48 (992 CZ gates)



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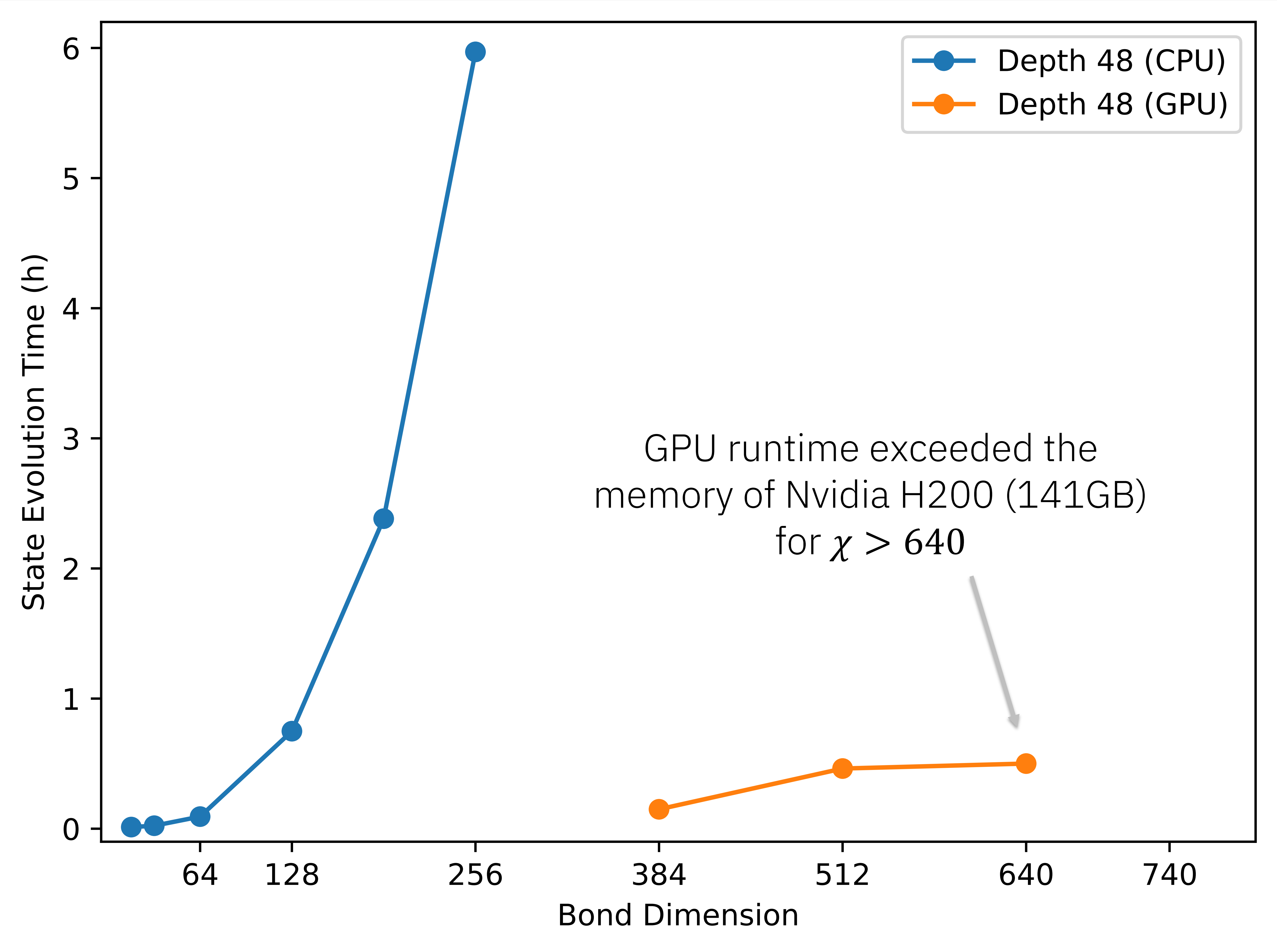


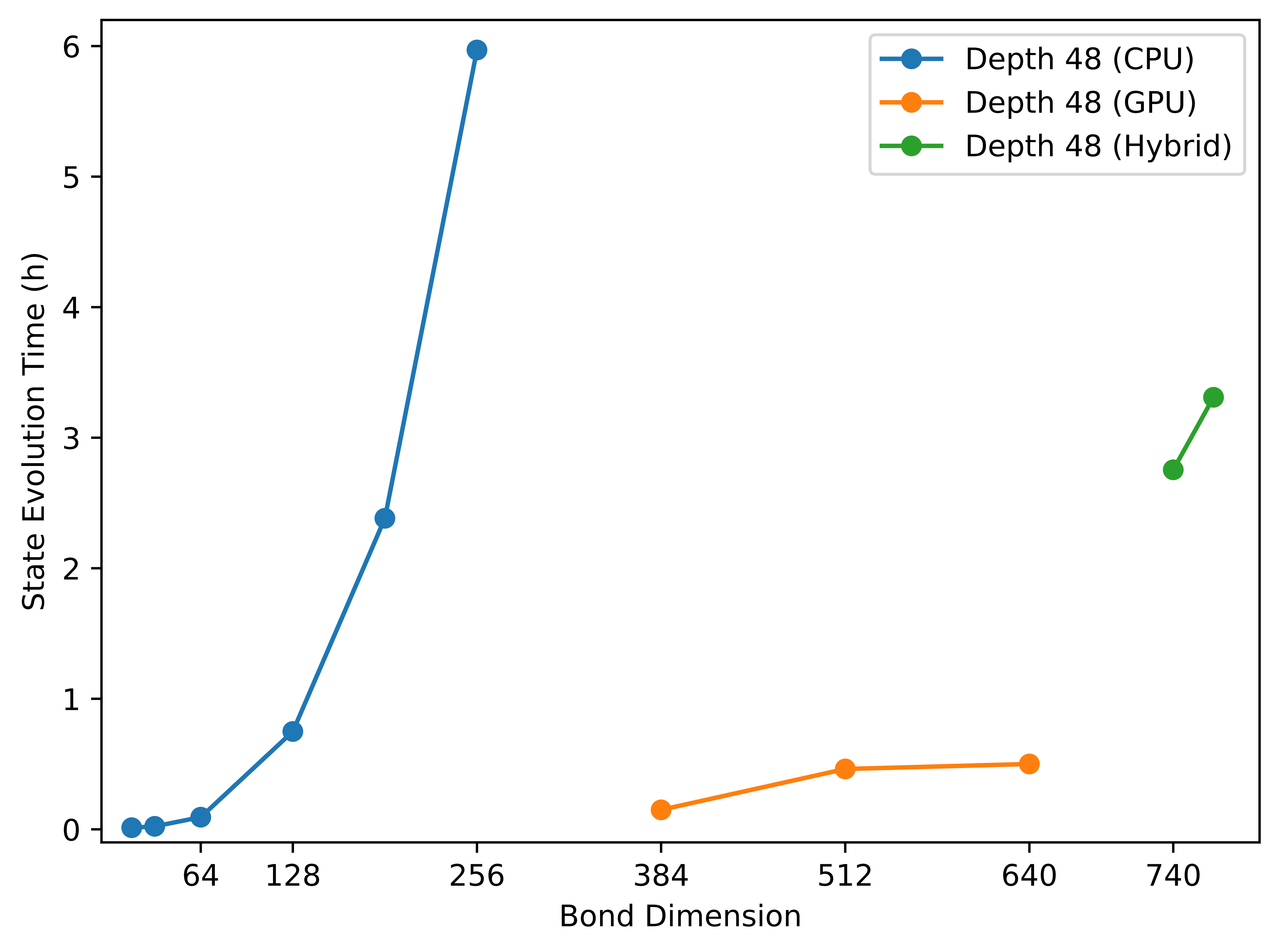
Example 3:
Observable Estimation

Increase resources for TN-BP and hope for convergence?

Hybrid GPU-CPU:
~ 3h to evolve
~ 1h to measure per observable per initial state

For reference:
Quantum runtime is ~ 1h
for *all* 512 initial states
for *all* diagonal observables

TN-BP runtime for OLE at Depth 48 (992 CZ gates)



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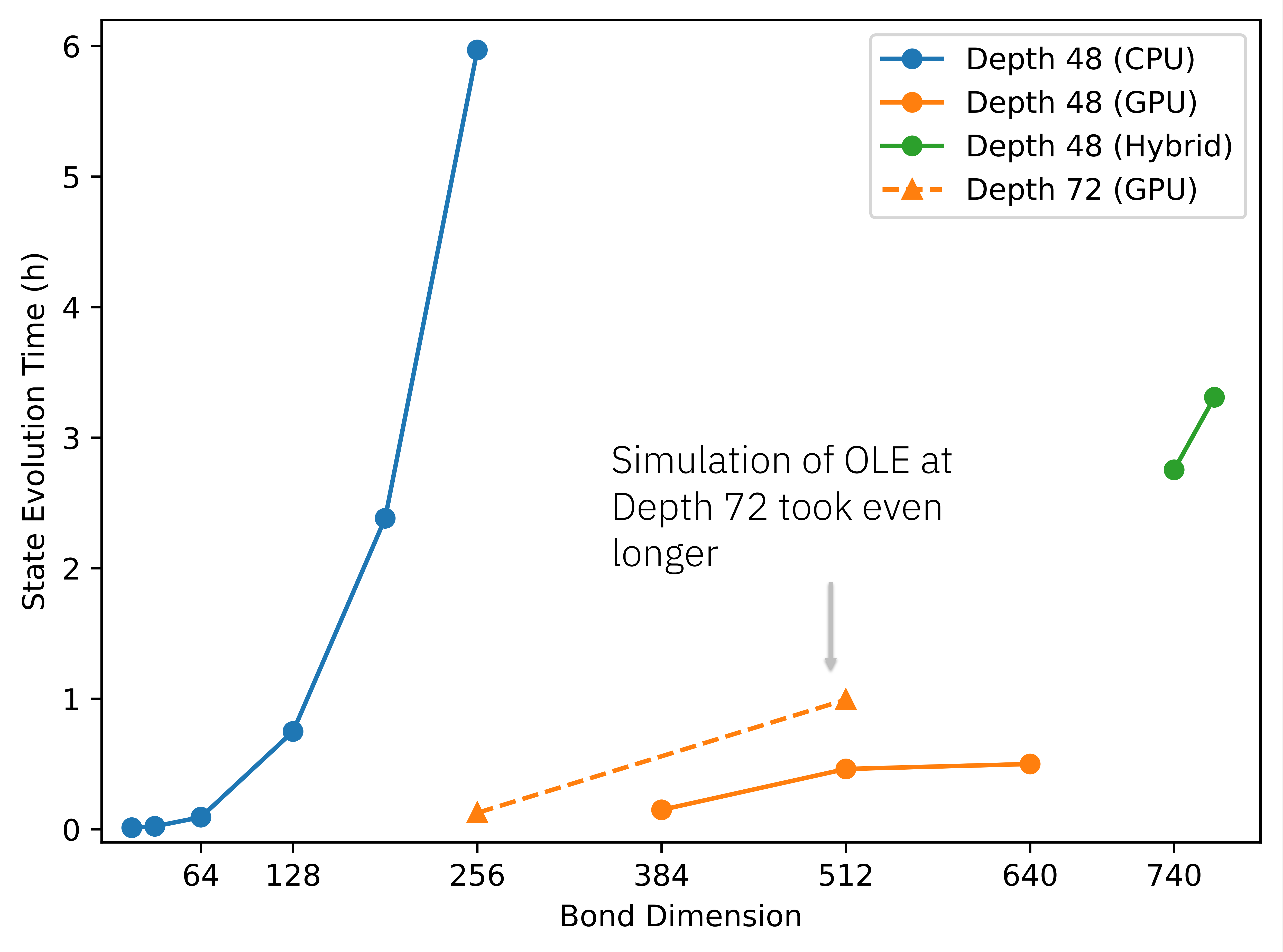

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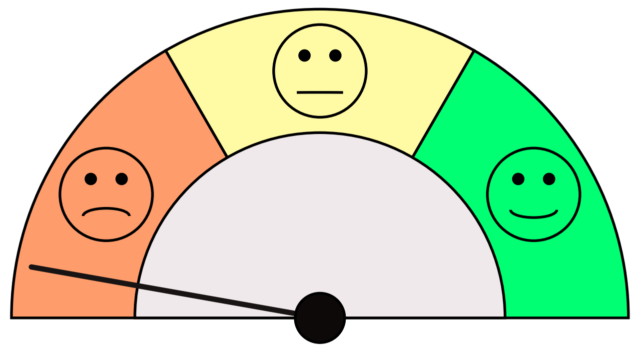
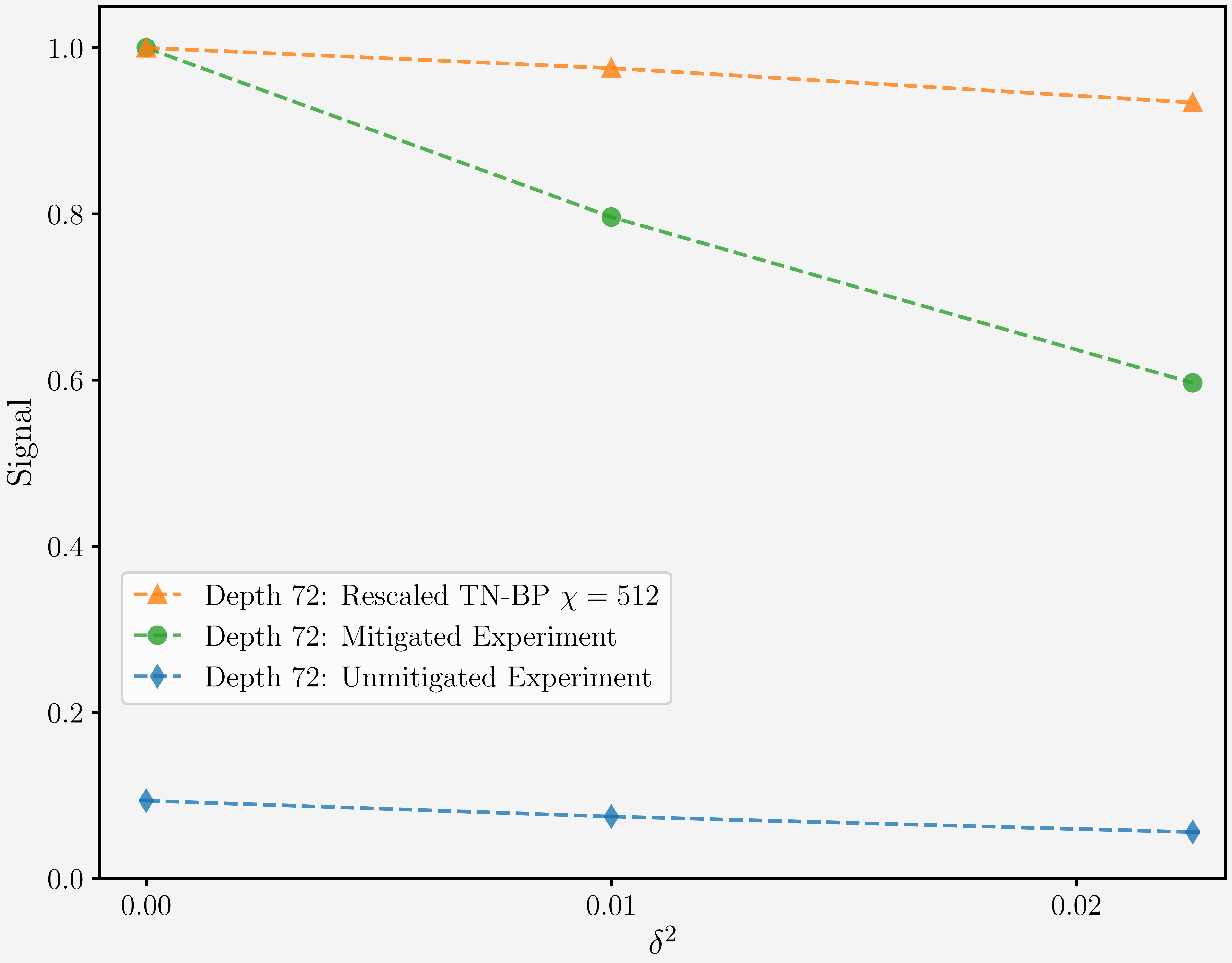


Trust-o-meter

OLE at Depth 72 (1488 CZ gates)



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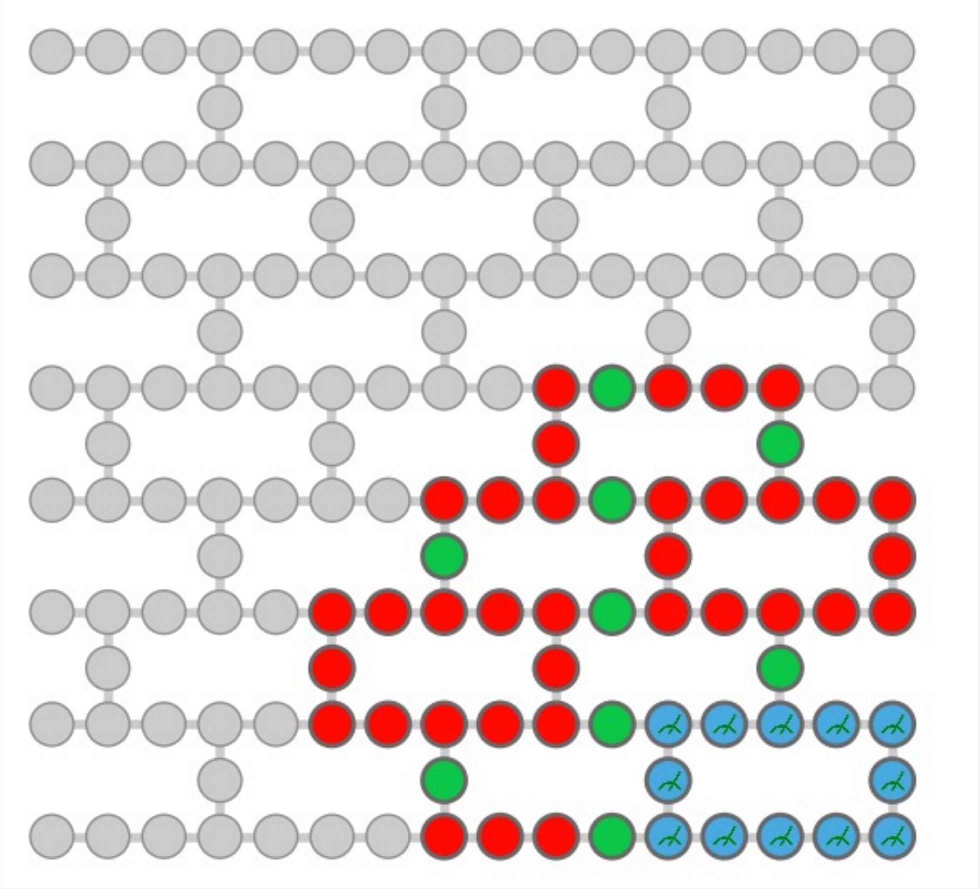
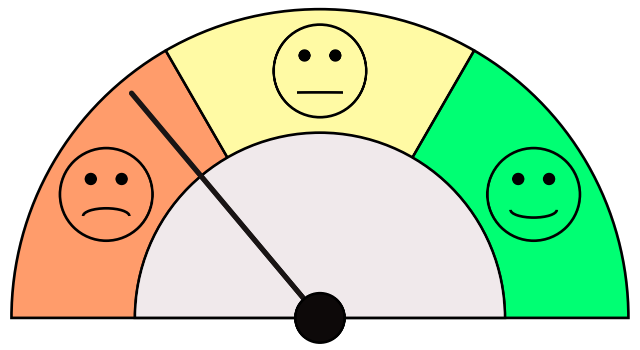


Example 3:
Observable Estimation

Example: Operator Loschmidt Echo

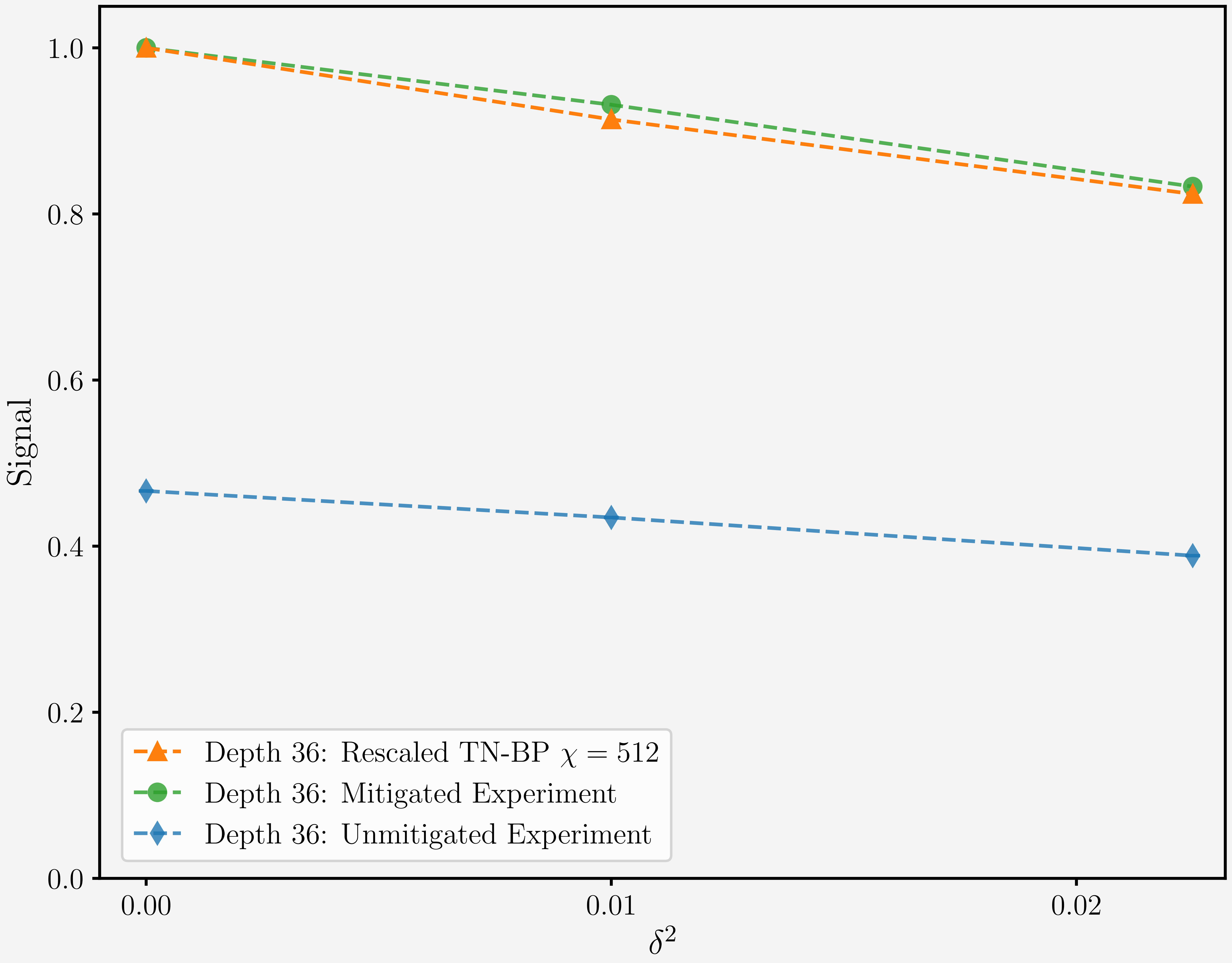
Heuristic error mitigation: Global Rescaling

Evidence 1: Mitigated quantum signal at Depth 36 agrees with the converged TN-BP simulation



Trust-o-meter

OLE at Depth 36 (744 CZ gates)



ibm_boston

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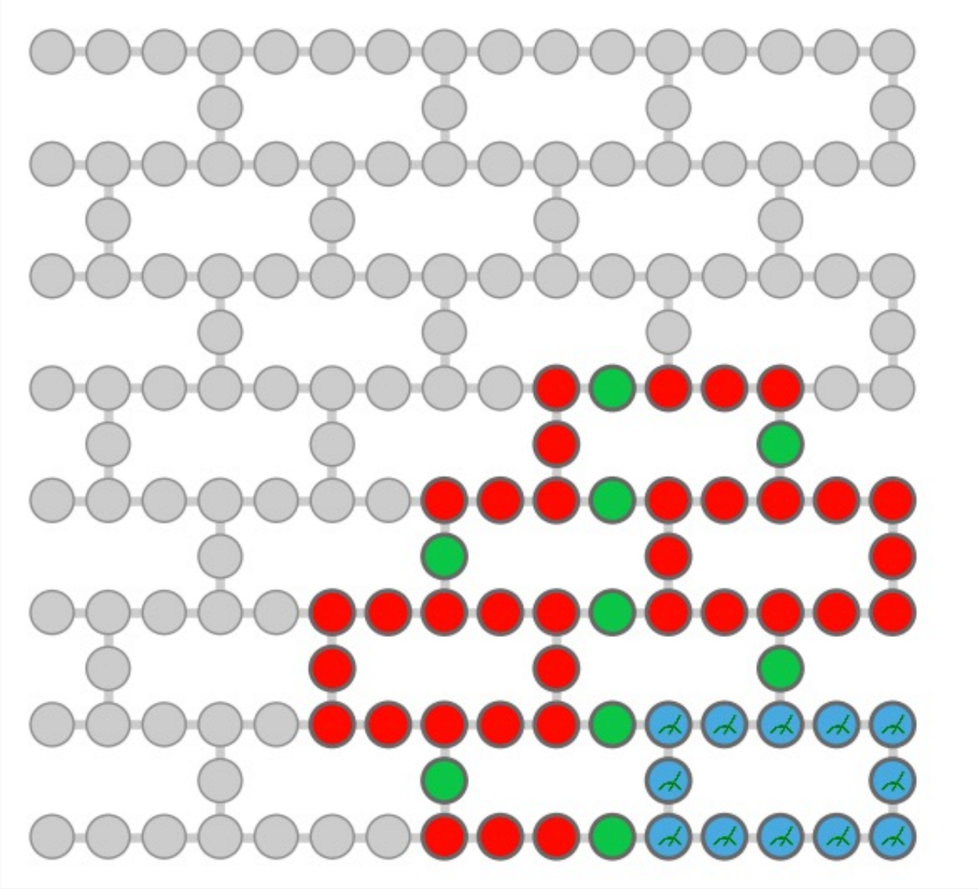
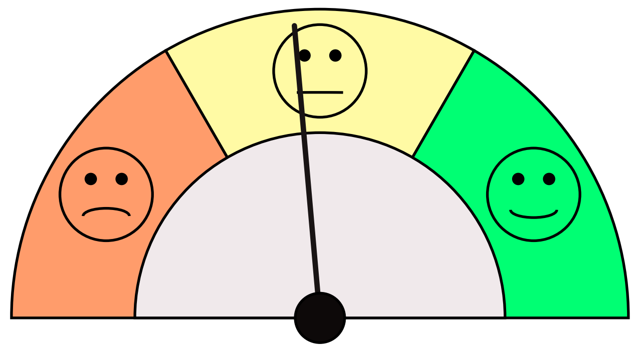


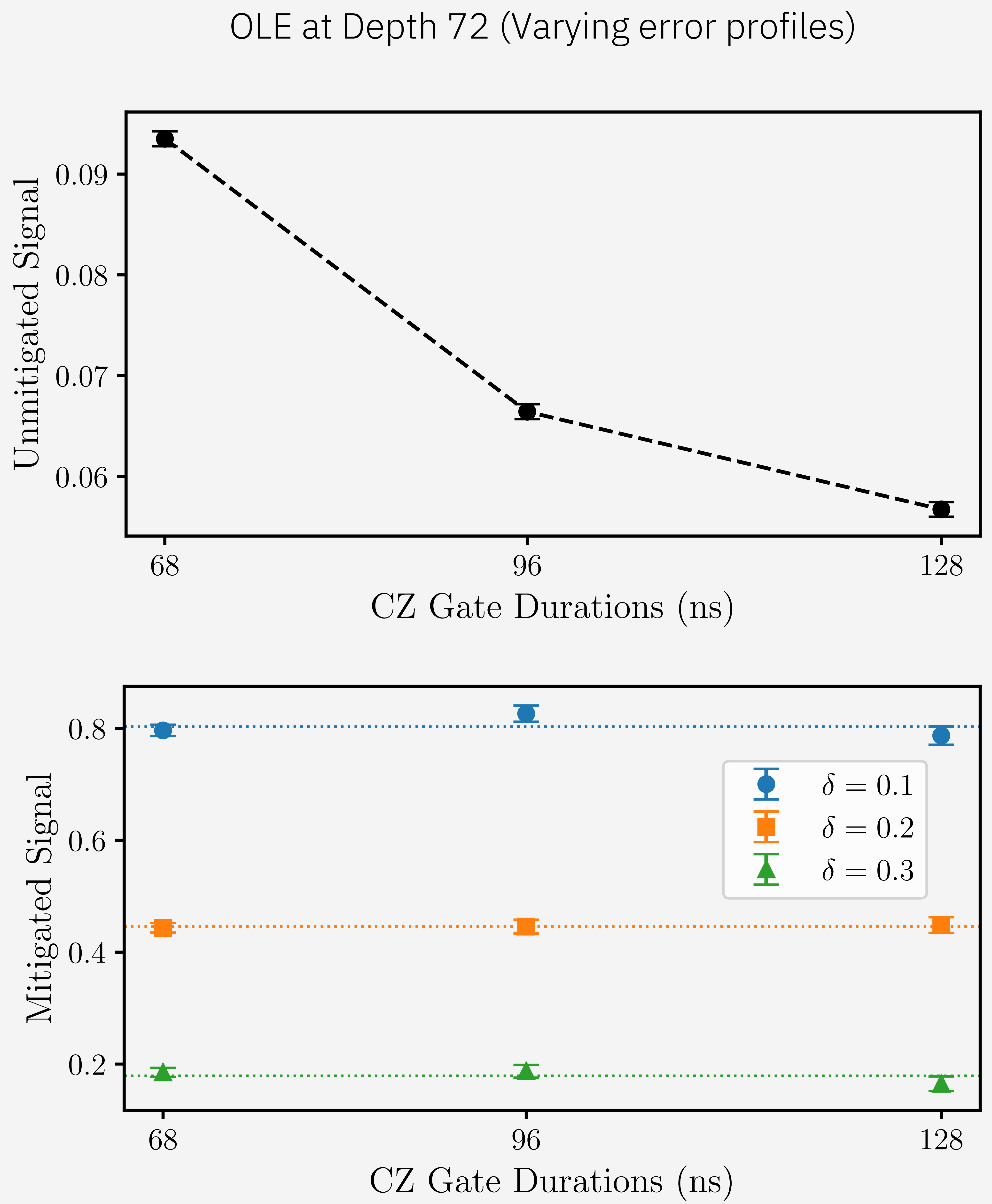
Example 3:
Observable Estimation

Example: Operator Loschmidt Echo

Heuristic error mitigation: Global Rescaling

Evidence 2: The mitigated quantum signal is reproducible under different experimental conditions





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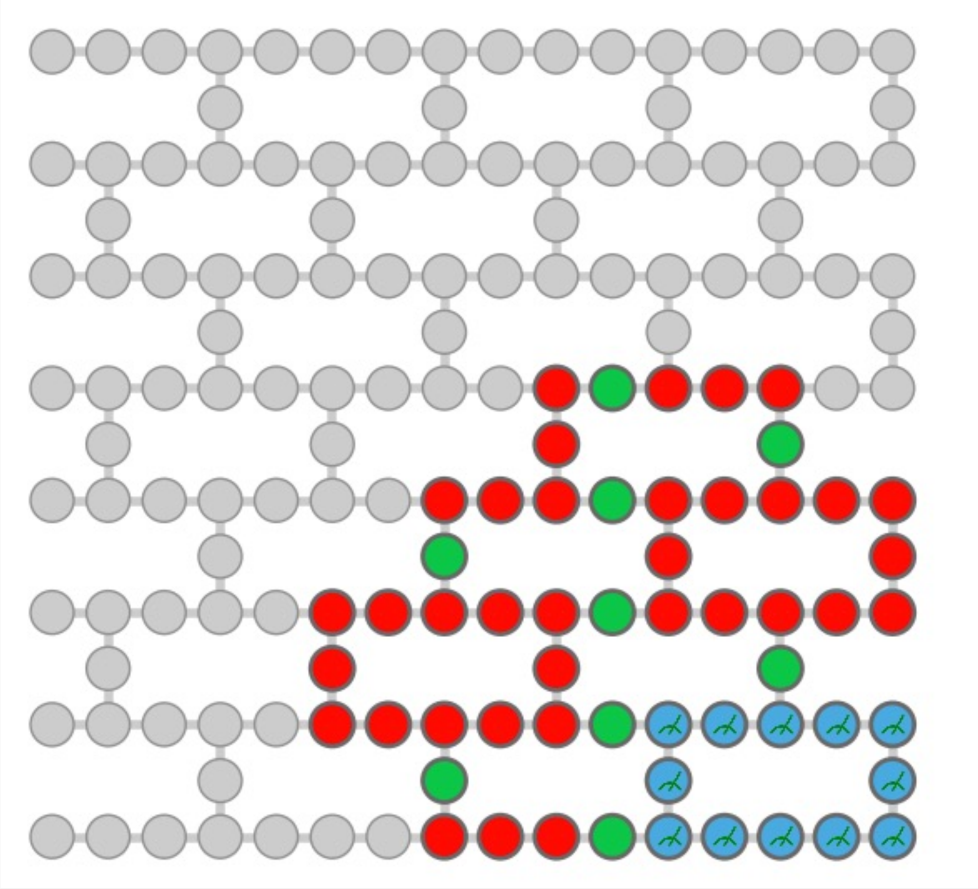
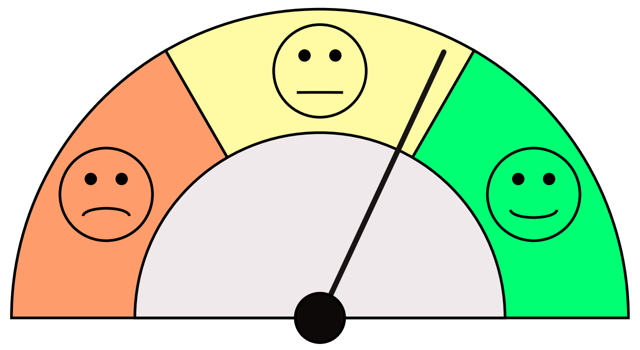


Example 3:
Observable Estimation

Example: Operator Loschmidt Echo

Heuristic error mitigation: Global Rescaling

Evidence 3: The mitigated quantum signal falls within a theoretically predicted range.



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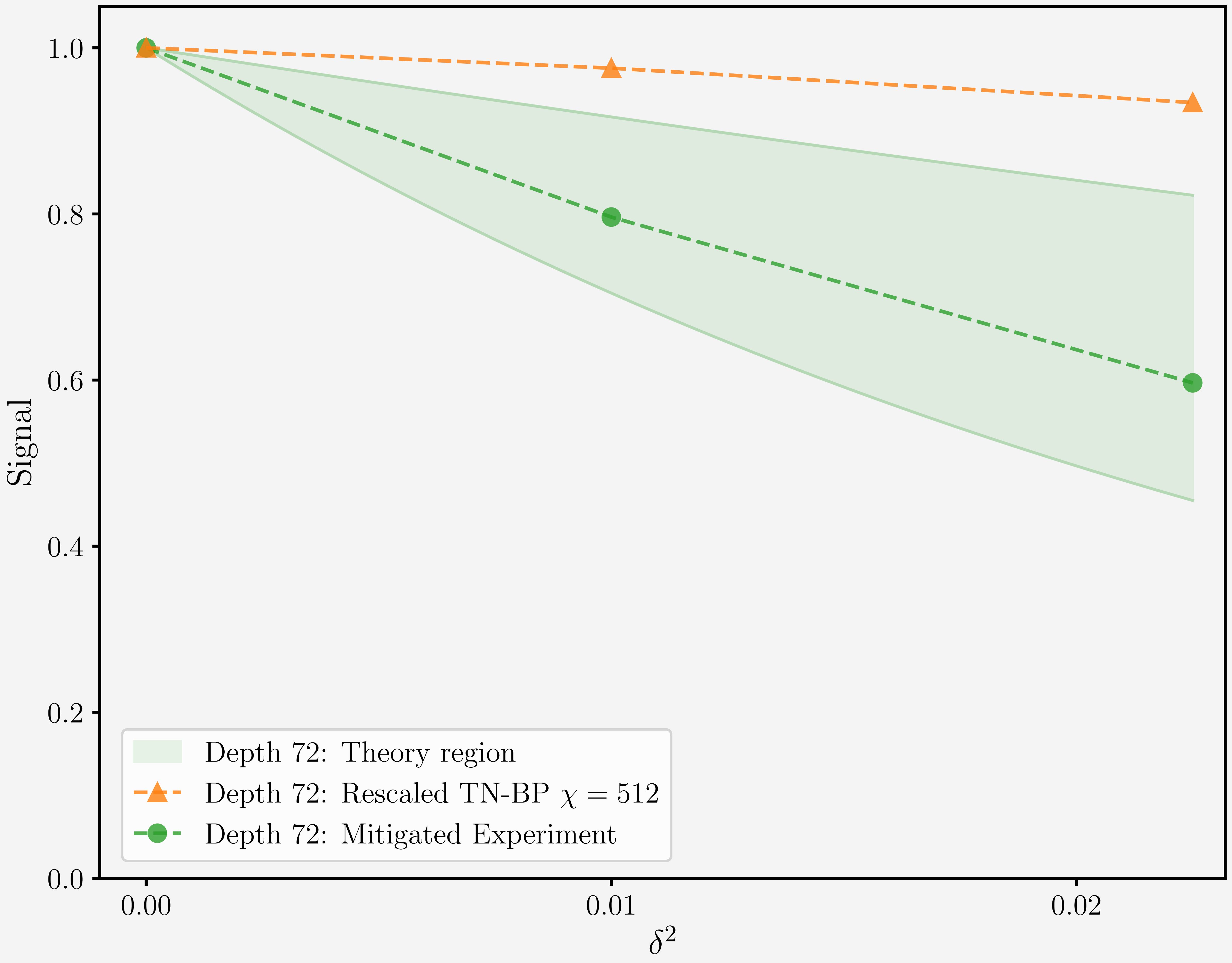
OLE at Depth 72 (1488 CZ gates)



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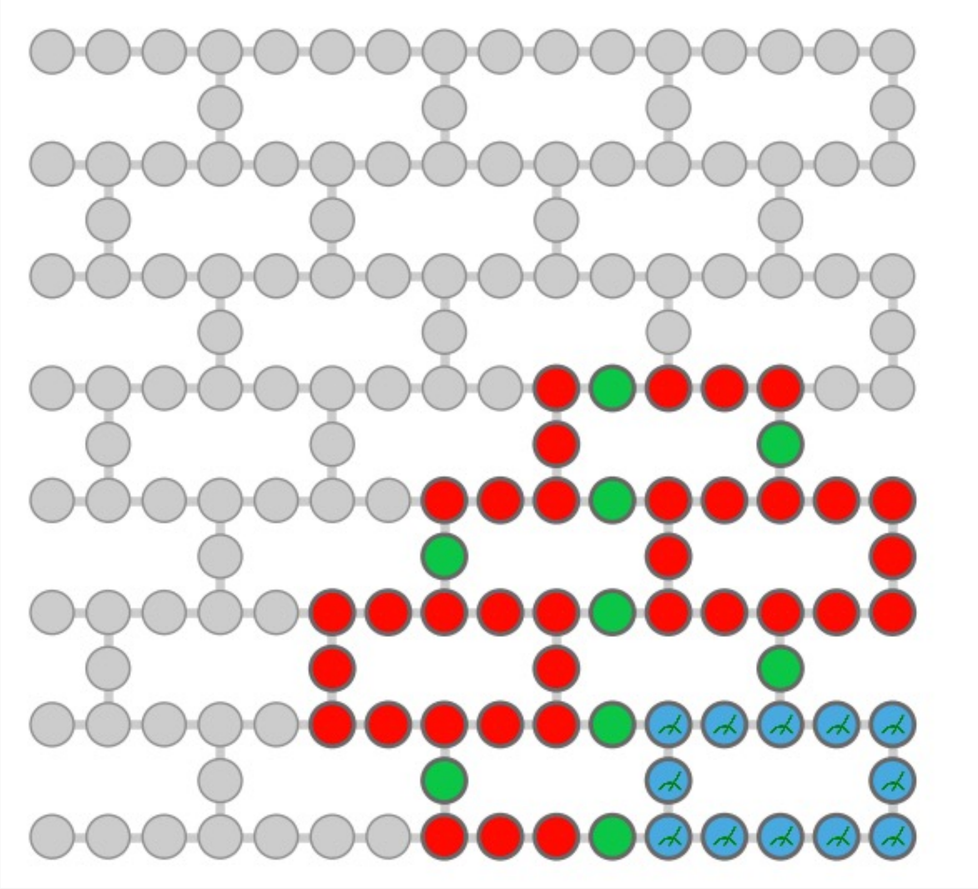

Example 3:
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Example: Operator Loschmidt Echo

Heuristic error mitigation: Global Rescaling

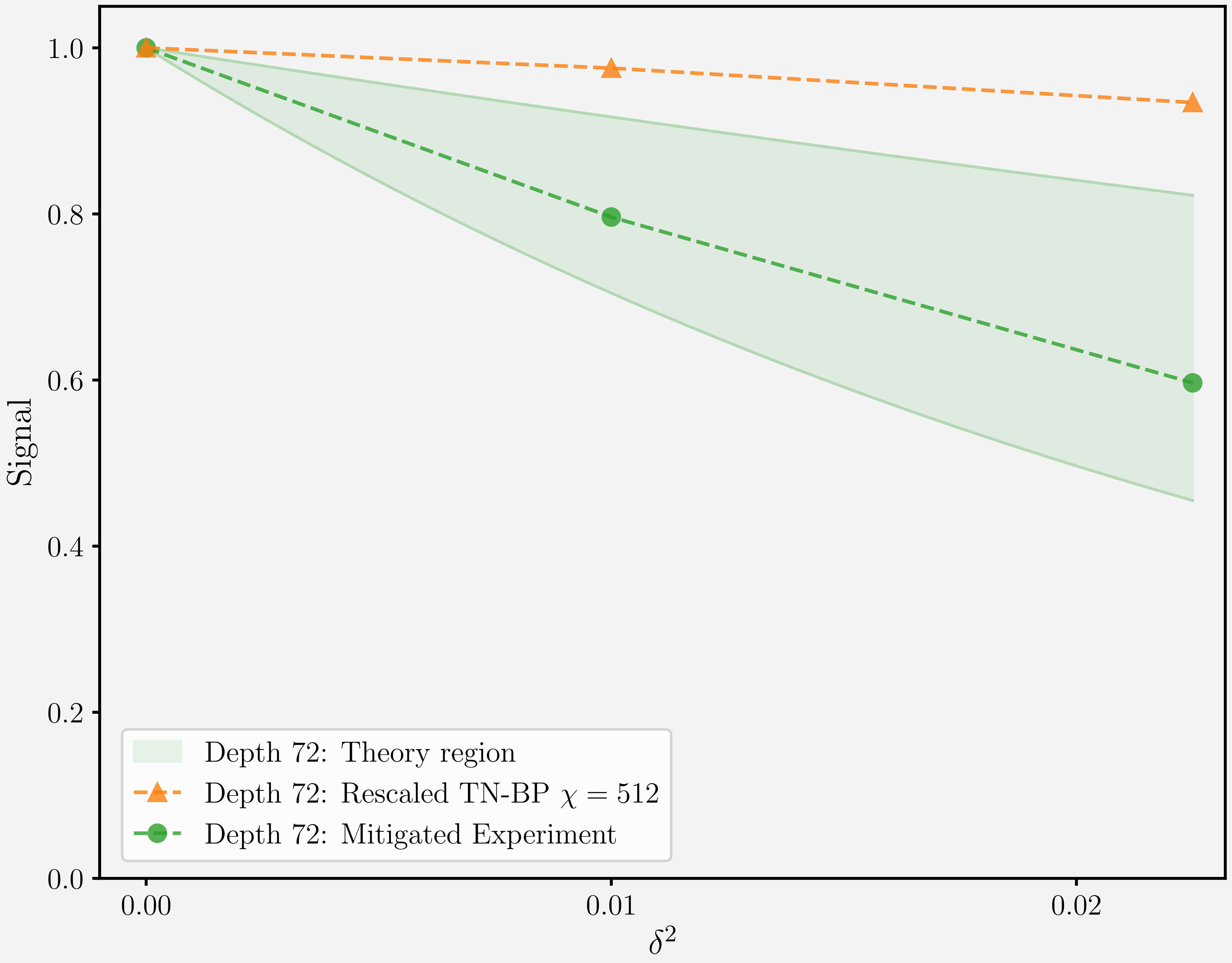
We can trust the answer from ibm_boston more than that of TN-BP

Quantum Advantage? No

- Benchmark against other classical heuristics?
- More trustworthy, but still not verifiable

Developing robust error mitigation and building trust on it is an important step towards QA

OLE at Depth 72 (1488 CZ gates)



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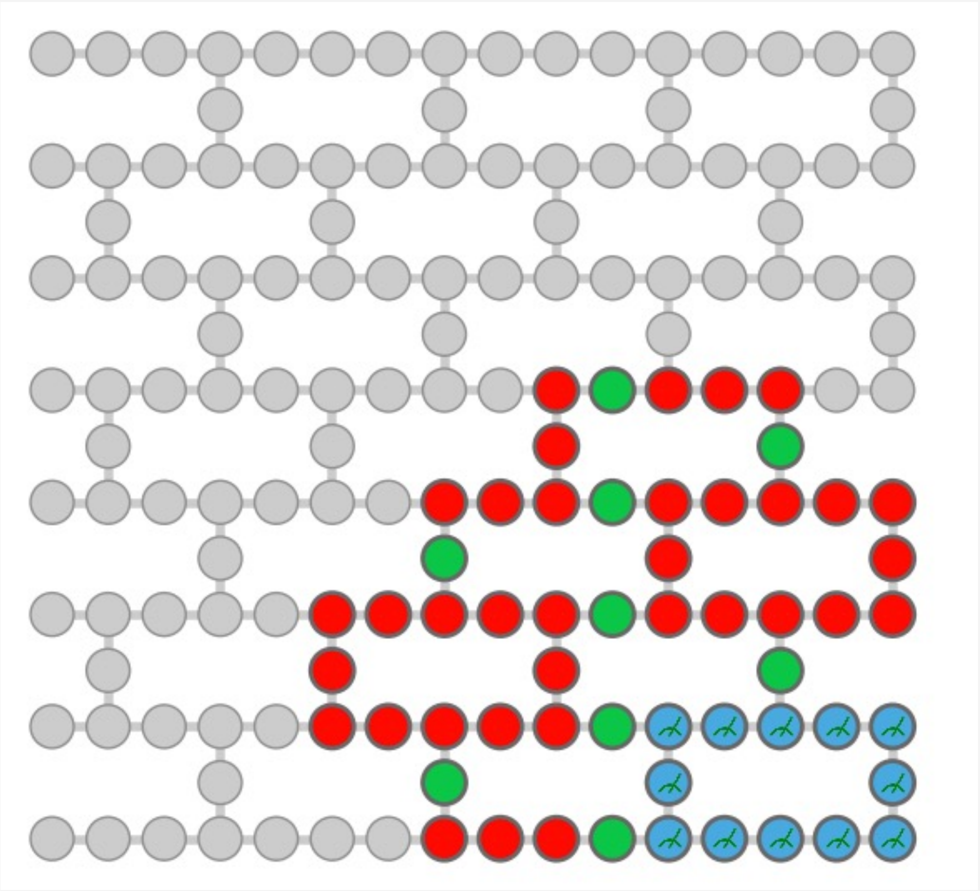

Quantum Advantage Tracker

Advantage trackers

Participate

Advantage trackers

Track verifiable quantum advantage candidates across three pathways, with current status of classical and quantum computations, supporting evidence and contributing organizations.

Observable estimations

Variational problems

Classically verifiable problems

Observable estimations 📊

Submissions in this tracker report expectation values for observables alongside rigorous error bars. Validation requires mathematically provable confidence intervals over the reported value.

View circuit options

Open submission ticket

Observable estimations 📊

Submissions in this tracker report expectation values for observables alongside rigorous error bars. Validation requires mathematically provable confidence intervals over the reported value.

View circuit instances

Open submission ticket

operator_loschmidt_echo

70×1872

Reset

Date ↓	Name / Institutions	Method	Circuit	Qubits	Gates	Expectation value [upper, lower bound]	Runtime (seconds)	Compute resources
2025-11-21	Single_path_MC_70×1872 **By:** University of Chicago, IBM, Algorithmiq, EPFL, Flatiron Institute	Single-path Monte Carlo	operator_loschmidt_echo_70×1872	70	1872	0.614 [N/A, N/A]	Q: - C: 1890	Q: - C: Apple M1 Max 64GB
2025-11-11	70×1872, ibm_boston **By:** IBM, Algorithmiq, Flatiron Institute	Global rescaling	operator_loschmidt_echo_70×1872	70	1872	0.673 [N/A, N/A]	Q: 4410 C: -	Q: ibm_boston C: -
2025-11-11	70×1872, BP-TN BD=512, rescaled **By:** Algorithmiq, Flatiron Institute	Belief Propagation TN, BD=512 rescaled by delta = 0.0	operator_loschmidt_echo_70×1872	70	1872	0.95247989 [N/A, N/A]	Q: - C: 20040	Q: - C: 1 x custo Platin
2025-11-11	70×1872, BP-TN BD=512, no rescaling **By:** Algorithmiq, Flatiron Institute	Belief Propagation TN, BD=512 no rescaling	operator_loschmidt_echo_70×1872	70	1872	0.873827508 [N/A, N/A]	Q: - C: 20040	Q: - C: 1 x custo Platin

Quantum Advantage Tracker

Quick, open, community-led review of candidates for quantum advantage and attempted solutions, both quantum and classical.

80+ submissions to date from 10+ organizations, including Algorithmiq, Blue Qubit, The Flatiron Institute, Oakridge National Lab, U. Maryland, U. Wisconsin Madison, U. Toronto, U. Chicago, Vector Institute, EPFL, RIKEN, IBM, Caltech, Los Alamos National Lab.

Different ways to participate:

- Have a candidate for quantum advantage? Submit a circuit instance
- Have a new classical simulation or want to demonstrate quantum capability? Submit a solution to an existing instance

Key takeaways

1

Quantum computation will soon be beyond the capability of reliable classical heuristics.

2

How do we trust any answers? Some problems have easily verifiable solutions.

3

For others, we do it the old-school way: collect evidence that the quantum computation is reliable.

4

Quantum Advantage is not a destination. It is the journey of building trust in quantum computation